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Reliable and Economical Aerodynamic Shape Optimization

Black-Box Gaussian-Process Bayesian Optimization with an Online-Selected Surrogate Prior

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Abstract

Aerodynamic shape optimization against a Reynolds averaged Navier Stokes solver costs minutes per evaluation, and a zoo of cheap estimators claims to predict the same physics. The natural idea is to turn that zoo into a gate: quantify each estimator's uncertainty and pay for the truth only where it declares itself uncertain. This thesis shows that the gate saves nothing, and not through a failure of calibration. The uncertainty does rank the error at the design the optimizer elects, but the gate trusts only designs already anchored by the solver, where trust is worth nothing, and the designs worth trusting are the uncertain ones. Verification is a necessity, not a setting.

At a fixed verification budget the economy comes from elsewhere. We measure three levers one at a time, the dimension of the search space, the acquisition, and the use of a cheap estimator as the prior mean of the Gaussian process rather than as a substitute for the truth. The prior is, with the reduction of the search space, one of the two largest levers: with a member suited to the case it reaches the target several times sooner. But its sign depends on which member is used, and a bad choice costs more than a good one earns, so selection is the condition under which the result can be used at all.

Criteria that judge a member by its accuracy away from the path of the optimizer fail, because closeness measured there does not survive the trip to where the optimizer goes. What works is to score each candidate on the *order* it puts on the designs the loop has already verified, online, in the spirit of ranking weighted ensembles, with one candidate standing for the absence of a prior. The selector most often recovers, without intervention, the member one would have named by hand, moves its weight to the absence of a prior when no member orders the anchors well, and retires by itself once the Gaussian process has enough data of its own.

The same machinery therefore fails as a gate and succeeds as a prior. Replacing a true evaluation requires the magnitude to be right where the optimizer is looking, which no learned surrogate delivers; guiding a search requires only the order to be right on the anchors, which it can.

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Chapter 1

Introduction

1.1 Aerodynamic shape optimization is expensive

Designing an airfoil means searching a space of shapes for the one that performs best at a given flight condition, and the only way to know how a shape performs is to solve the flow around it. A Reynolds averaged Navier Stokes (RANS) solution takes minutes on the hardware used for this thesis. A single optimization is therefore most of a day of computation, and a design study that sweeps several conditions is a week. That cost is the reason shape optimization is not run the way a textbook would suggest, by evaluating every candidate the optimizer proposes.

The cost also decides what counts as progress. Making the solver faster is one route, and it is not ours. The other is to spend fewer solutions for the same design, which is a question about the optimizer rather than about the physics. This thesis takes the second route, and its unit throughout is the number of high fidelity evaluations spent, never the wall clock.

This work was carried out at the Computer Vision Laboratory of EPFL, which works on learned models of aerodynamic behaviour and on their use inside shape optimization. The subject started from a practical observation. A RANS evaluation is too slow to sit inside an iterative optimizer, cheap learned surrogates such as NeuralFoil approximate it well but fail without warning once the search leaves the region they were trained on, and when several estimators are available, choosing the one with the best ratio of performance to cost is far from obvious. The initial question was therefore whether quantifying the uncertainty of these estimators would make an optimization both reliable and economical: reliable because the optimizer would know when not to believe a surrogate, economical because it would pay for the solver only then. Aerodynamics suits that question because the truth exists and has a price, one RANS evaluation, so whether an uncertainty keeps its promise can be measured rather than assumed. The answer this thesis gives is not the one expected at the outset, and Section 1.2 explains why.

1.2 A zoo of estimators, and the question of choosing between them

A cheap estimator answers in milliseconds where the solver takes minutes, and a decade of aerodynamic machine learning has produced many of them, trained on different data, at different fidelities, for different flow regimes. None is as accurate as the solver and none is accurate everywhere. That plurality is the starting point of this thesis and the origin of its title. If no single estimator can be trusted throughout, perhaps the right one can be chosen at each point of the search, and perhaps that choice can be made to pay.

Choosing requires a currency, and the natural one is the error each estimator would make against the reference that will judge the design. That error is unknown at decision time, so it is replaced by an estimate of itself, obtained in black box mode by measuring every member on a common held out set. The optimizer then reads that estimate twice. It selects, at each candidate, the member whose uncertainty is smallest. And it decides, at each candidate, whether that uncertainty is small enough

to believe the member outright or whether the solver must be called. Selection and economy come from the same quantity, and a solution is saved every time the optimizer agrees to believe. That is the bet, it is a reasonable one, and testing it is the first half of this thesis.

1.3 Reliability is lowest where the optimizer goes

The bet has a weak point that its statement hides. An optimizer does not sample its space, it takes the maximum of an estimate. Among candidates of comparable true value it retains the one whose error happens to be favourable, so the design it elects is selected for the optimism of the estimator and not only for its quality. The consequence is old and has a name, the optimizer's curse [41]. It means that a calibration established on a population says little about the subpopulation an optimizer delivers, and it is measured here rather than assumed: the same surrogate is honest on held out airfoils and markedly optimistic on the designs it elected, with nothing changed in the model between the two measurements.

What follows is the result that redirected this thesis. The selection half of the bet turns out to add little over using a single good member, provided one knows in advance which member that is. The economy half fails outright, and it fails for a reason no recalibration repairs. Where it would be worth believing an estimator the error is large and the uncertainty says so, so the gate refuses and pays. Where the gate agrees to believe, the candidate sits in a region already anchored by earlier solutions and believing it changes nothing at best; at worst the beliefs it lets in pull the search toward phantoms and the loop ends below verifying everything. Trust is granted exactly where it is worthless. Refuting the bet cleanly, on many cases and with the mechanism identified, is the first contribution of this thesis.

A negative result of that kind is only useful if it sharpens the question, and it does. If uncertainty cannot buy evaluations by skipping them, then the economy an expensive optimization needs must come from somewhere else, and the second half of this thesis asks where. The answer returns the zoo to the argument in a different role. The same estimators that cannot replace a solution can supply the shape the optimizer follows, as the prior mean of the process, with every design still verified. Selection did not disappear from the problem, it changed what it selects for.

1.4 Contributions and outline

The thesis follows that arc. We build a pool of aerodynamic estimators and a black box uncertainty model for each against RANS, we test the bet on nine optimization cases spanning five flow regimes and two objectives, we find it refuted, and we then measure where the economy actually comes from.

- A uniform black box measurement of a pool of aerodynamic estimators against a single reference, regime by regime, with an uncertainty model calibrated in distribution on held out airfoils.
- A refutation of the uncertainty gate across the nine cases, with the selection effect that causes it identified and measured, rather than a miscalibration blamed for it.
- A measurement of the levers that do economise, the search space, the acquisition and the surrogate used as the prior mean of the process, each varied one at a time against the same reference loop.
- A way to choose that prior online among the zoo, by the order each member puts on the designs the loop has already verified, with the option of choosing no prior at all, so that the economy does not depend on naming the right member in advance.
- The distinction that follows, and it is the sentence we would keep if we kept one. Replacing a true evaluation requires the magnitude of a surrogate to be right where the optimizer is looking,

which no learned surrogate delivers; guiding a search requires only its order to be right, which it can.

Chapter 2 introduces only the machinery the argument needs, the estimators, the data and the probe, the shape representation, the optimizer and the uncertainty model. Chapter 3 states the problem, the single property the uncertainty must satisfy for the bet to work, the three hypotheses we test and the protocol. Chapter 4 refutes the gate and explains why. Chapter 5 measures the three levers that remain and ends on how to choose the prior. Chapter 6 states what the thesis shows, what the uncertainty is actually for, the limits of the study, and what the result makes possible.

Chapter 2

Background: the toolbox

This chapter introduces only the machinery the argument relies on, and it is organised around three needs. We need an estimator that is cheap enough to be queried thousands of times, we need a way to know when that estimator is wrong with respect to the reference we will be judged against, and we need an optimizer able to exploit both. The zoo and the data answer the first need, the probe set and an input output conditioned uncertainty model, IO-CUE, the second, the shape representation and the Bayesian optimizer the third. Every number quoted here is read from a table or a figure built from our own data.

2.1 A zoo of aerodynamic estimators

Three families of estimator span the cost against reliability axis, and reliability is always measured in one way in this thesis, as agreement with a high fidelity reference. At the top sits the Reynolds averaged Navier Stokes solver SU2 [13], which resolves the flow field and stays trustworthy across regimes at a cost of minutes per evaluation. We treat it as ground truth and we validated our setup against the classical NACA 0012 measurements [19]. In the middle, XFOIL couples a panel method with an integral boundary layer [12] and returns a polar in milliseconds, but its convergence degrades near stall and its accuracy falls in transonic flow [2]. At the bottom, NeuralFoil [40] is a neural network that imitates XFOIL and answers almost instantly. This last point matters for everything that follows. NeuralFoil was trained on XFOIL, not on RANS, so it can be confident and still be far from the reference.

Rather than commit to one estimator we assemble a zoo of seven members, and the way it is built matters as much as its size. Two are not ours, NeuralFoil, introduced above, and AirfoilANN, a published network trained on a collection we did not assemble. XFOIL itself is not a member: it is the teacher of NeuralFoil, and it would bring a second copy of the same source rather than an independent one, at a cost several hundred times that of the networks. The five others are networks we trained on the harmonised dataset of Section 2.2, and they differ only by the rows they were shown. One is the *global* network, trained on every source pooled. Two are *regime experts*, each trained on the rows of a single regime of Table 2.1, turbulent attached and subsonic compressible. Two are *source experts*, trained on the rows of a single large provider so that their bias against the reference is at least internally consistent, one on the UniFoil collection and one on the HAM2D sweeps. This partition is the object of the thesis in miniature. It spans the two ways a cheap model can be specialised, by flow regime and by data provenance, and it lets the question of Chapter 4 be asked properly, namely whether an uncertainty estimate can pick the right one at the right place. The global network is a multilayer perceptron with three hidden layers, rectified linear activations and light dropout. It takes the twenty two coefficients of the class shape transformation (CST) of Section 2.3, the flow conditions (Mach, Re, α) and two geometric descriptors, the thickness and the camber, and it predicts the lift and drag coefficients C_L and C_D against the RANS reference. Drag is quoted throughout in counts, one count being 10^{-4} of C_D. It is trained on the training split of every source of Table 2.1 pooled together, the OVERFLOW set excluded, so it has seen all six regimes but none of the probe airfoils. The regime

experts share the same architecture and are trained on the rows of a single regime. Figure 2.1 places these members on the cost against reliability map, with reliability read on a held out probe against the RANS reference. Two facts stand out. The global network is the most reliable cheap member on average, and every cheap member is unreliable somewhere, each in a different regime. The first fact is partly built in. The global network learned from RANS data of the same sources, on other airfoils, whereas NeuralFoil learned from XFOIL, so the comparison rewards the estimator whose training fidelity matches the reference. Far from being a flaw, this is the point. The fidelity a model was trained against bounds the fidelity it can deliver, and it is also why the best prior, in Chapter 5, will not be the most accurate member. The second fact is visible regime by regime. NeuralFoil drifts in transonic flow and its drag error is an order of magnitude above the global network in attached flow. A narrow expert degrades outside its regime, both regime experts reach several hundred counts in post stall flow, which is why experts never appear alone in the zoo. Specialisation pays only where there is data to specialise on, and Figure 2.1 makes that clear. Each regime expert is the best member of its own regime, and both of those regimes are densely sampled; in the sparsely sampled regimes, near stall and post stall, the best member is the generalist trained on everything. A regime expert is therefore a better model of its regime only when that regime carries enough data, and specialisation and data compete.

All members are treated as black boxes. The physics solvers expose no gradients, the neural surrogates differ in architecture and cannot be retrained, and for an external model we know neither its training data nor the fidelity of that data. The method therefore works with any estimator that predicts the same physics, including a closed commercial model, as long as we can query it and compare it to the reference. We deliberately gathered cases across regimes where XFOIL and NeuralFoil are known to degrade, not only where they shine. That is what gives meaning to the words model selection in the title, and whether such selection actually helps on the cases we test is a question this thesis answers rather than assumes.

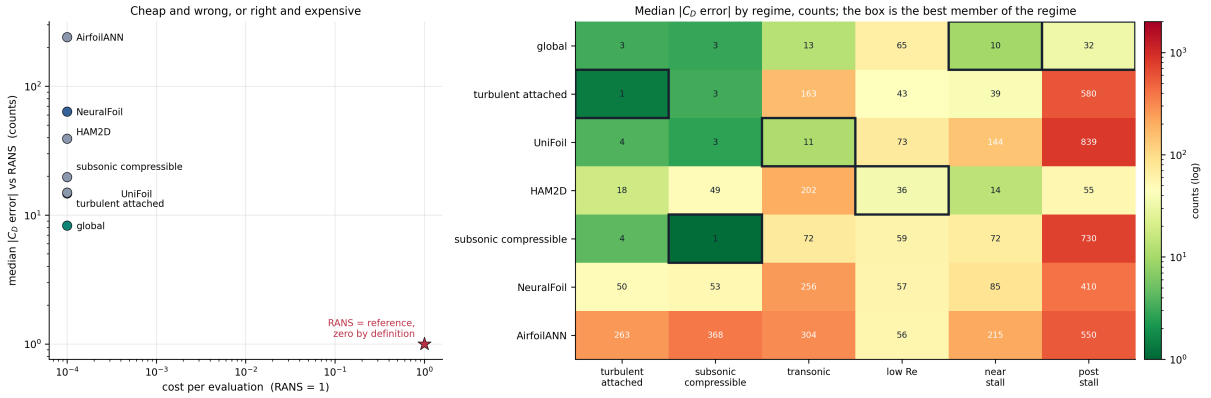


Figure 2.1: The estimator zoo judged against the RANS reference on the held out probe of Section 2.2. *Left:* cost per evaluation, normalised by one RANS solve, against the median absolute drag error, both on logarithmic axes. *Right:* the same drag error regime by regime, in counts, with the best member of each regime boxed. A regime expert wins inside its own regime where that regime is densely sampled, and is beaten by the generalist where it is not. No member is good everywhere. Reliability here is agreement with RANS, not a model's own confidence.

2.2 A harmonised multi source dataset and a free probe set

Training and judging the zoo requires broad and trustworthy coverage, and no single public dataset provides it. Existing collections concentrate on incompressible, fully turbulent flow, AirfRANS being a typical example [7], and they miss the transition, stall, and shock physics that an optimizer explores. We therefore built a consolidated dataset from seven open source RANS collections issued by five

providers, listed in Table 2.1. Each brings a piece of the envelope that the others lack. UniFoil [23] supplies the bulk, tens of thousands of airfoils from low subsonic to transonic Mach numbers, in fully turbulent and in natural laminar flow variants, so it carries the compressibility and the transition physics. The HAM2D sets [34, 35] sweep the incidence up to twenty degrees at a fixed Mach number, so they carry near stall and post stall behaviour that no other source has. The Agarwal set [3] works at a single low Reynolds number, so it carries the low Reynolds regime. Two small sets, an AIAA workshop case and an OVERFLOW set kept strictly as an external test, add a second solver family. Some sources are thin, a single Mach number or a single Reynolds number, and we say so in the table rather than hide it. The result is not an incompressible dataset. Mach reaches the transonic range, the Reynolds number spans two decades, the incidence reaches stall, and six flow regimes are represented.

Table 2.1: The seven open source RANS collections, from five providers, harmonised into the dataset. Ranges are read from the data. Regime labels: TA turbulent attached, SC subsonic compressible, TR transonic, LR low Reynolds, NS near stall, PS post stall. The OVERFLOW set is never used for training.

Source	Solver	Rows	Airfoils	Mach	Re	α [deg]	Regimes
UniFoil turbulent [23]	ADflow	493 574	39 998	0.10 to 0.85	10^6 to 10^7	−2 to 6	SC, TA, TR
UniFoil laminar [23]	ADflow	84 642	6716	0.10 to 0.85	10^6 to 10^7	−2 to 6	SC, TA, TR
HAM2D sweep [34]	HAM2D	110 110	1735	0.10	3×10^6 to 9×10^6	−4 to 20	TA, NS, PS
HAM2D high Re [35]	HAM2D	17 992	8996	0.10	9×10^6	4 to 12	TA
Agarwal [3]	OpenFOAM	33 705	2946	0	10^5	−4 to 8	LR
AIAA case 3	published	597	597	0.20 to 0.75	10^6 to 6.5×10^6	−3 to 5	SC, TA, TR
OVERFLOW (test only)	OVERFLOW	1600	16	0.25 to 0.90	7.5×10^4 to 8×10^6	−20 to 20	all six
Total		742 220	61 004	0 to 0.90	7.5×10^4 to 10^7	−20 to 20	six

The harmonisation is not cosmetic. All sources are mapped onto one schema that associates a twenty two coefficient CST shape and the flow conditions, Mach number, Reynolds number and incidence, with the lift, drag and pitching moment coefficients C_L, C_D and C_M. Geometries that came in other parameterisations are refitted to the common CST basis. Sources are tagged by solver because different solvers disagree in the same regime, and the tag is kept so that no disagreement is silently averaged away. Most important, the split between training, validation, and test is made by airfoil and not by row. Every polar of a given geometry lands in a single split, so no shape seen in training can reappear at evaluation. Skipping this step makes any uncertainty estimate look far more reliable than it is.

The same data gives us, for free, the probe set on which every zoo member is judged. Its RANS labels already exist in the sources, so it costs no new solve. It is drawn entirely from the validation split, it is stratified by regime so that the three dense regimes do not drown the rare ones, and it keeps thousands of distinct airfoils in each dense regime, Table 2.2. Because it is held out by airfoil, the error we measure on it for a network we trained ourselves is genuine out of sample error. We verified the guarantee directly. The airfoils of the probe overlap the training airfoils of our networks in zero cases. The validation split did play one role during training, it chose the stopping epoch of each network, without ever entering a gradient step. We state it because it is the only channel through which the probe could have influenced a model, and it is a weak one. The probe is split once more into a fitting part and a calibration part, and the calibration part is present in every regime, so the calibration of the uncertainty model in Section 2.5 is checked regime by regime and not only on average. One honest limit remains. The near stall and post stall regimes rest on a few hundred airfoils, so conclusions drawn there are weaker than in the dense regimes.

The probe is labelled by the solvers of the sources, not by the solver that defines truth inside the loop. A second, smaller set therefore judges the zoo against our own solver directly: 1261 evaluations over 309 airfoils of the probe, recomputed with our SU2 across five of the regimes and two further sampling strata, keeping only runs whose drag had settled within two per cent. It is the set Section 2.5 reads the zoo on.

Table 2.2: Composition of the held out probe set. All airfoils belong to the validation split and none appears in the training split of any network we trained.

Regime	Airfoils	Evaluations	of which calibration
turbulent attached	3638	6000	1878
subsonic compressible	4125	6000	1800
transonic	3970	6000	1857
low Reynolds	443	5039	1517
near stall	232	3465	1099
post stall	134	1350	399
Total	7468	27 854	8550

2.3 Shape representation: CST and a POD latent space

Airfoils are described with the class shape transformation of Kulfan [26]. Each surface is the product of a class function, which fixes the round leading edge and the sharp trailing edge, and a shape function expanded in Bernstein polynomials of degree ten,

$$y^{\pm}(\psi) \;=\; \psi^{1/2}(1-\psi)\sum_{i=0}^{10} A_i^{\pm}\binom{10}{i}\psi^{i}(1-\psi)^{10-i}, \qquad \psi = x/c \in [0,1],$$

that is eleven Bernstein coefficients per surface and a trailing edge of zero thickness, twenty two parameters in total. This is the vector $a \in \mathbb{R}^{22}$ shared by every estimator and by the RANS mesh generator.

Real airfoils do not fill this space. They lie close to a low dimensional manifold, and a proper orthogonal decomposition (POD) of the CST vectors of the dataset makes that explicit [33]. With $\bar{a}$ the mean shape and Φ_k the first k modes,

$$a \;\approx\; \bar{a} + \Phi_k z, \qquad z \in \mathbb{R}^k, \qquad \frac{\sum_{i\le k}\lambda_i}{\sum_i \lambda_i} \text{ is the energy captured,}$$

where λ_i are the eigenvalues of the covariance of the CST vectors, in decreasing order. Figure 2.2 shows that eight modes capture ninety nine per cent of the variance of the shapes. That figure is measured on the coefficients, and the right panel says what it means on the shape itself. Eight modes leave a residual surface error of about one per cent of the chord on the airfoils of the dataset, and about twice that on the designs the free search delivers. The reduction is therefore a restriction and not a lossless reparametrisation. Whether the restriction costs performance is an empirical question, and Chapter 5 answers it by running the same optimizer in a and in z.

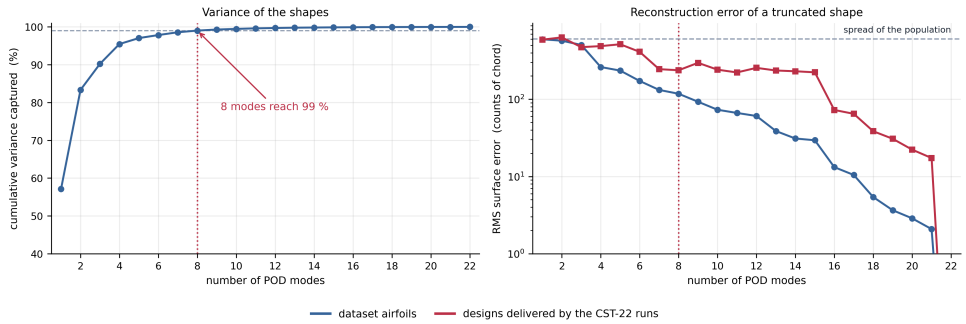


Figure 2.2: *Left:* cumulative variance of the airfoil shapes captured by the POD modes of the CST coefficients; eight modes reach ninety nine per cent. *Right:* what that leaves on the shape itself, the root mean square surface error of a shape truncated to k modes, in counts of chord, one count being 10^{-4} of the chord, for the airfoils the decomposition was fitted on and for the designs the free twenty two coefficient runs delivered. The truncation is not free, and it costs more on the delivered designs than on the airfoils of the dataset.

2.4 Black box Bayesian optimization

We optimize with Gaussian process Bayesian optimization [22, 39], which suits an expensive objective without gradients. In the engineering literature the same model is called kriging, and we use the two words as synonyms. A Gaussian process (GP) $f \sim \mathcal{GP}(m, k)$ with prior mean m and kernel k, conditioned on n evaluations y at points X_n, has posterior mean and variance

$$\mu_n(x) = m(x) + k_n(x)^\top \big(K_n + \sigma_\varepsilon^2 I\big)^{-1}\big(y - m(X_n)\big), \qquad s_n^2(x) = k(x,x) - k_n(x)^\top \big(K_n + \sigma_\varepsilon^2 I\big)^{-1} k_n(x),$$

and an acquisition function proposes the next design, $x_{n+1} = \arg\max_x \alpha(x; \mu_n, s_n)$. The choice of the acquisition matters less than the literature once suggested. Expected improvement, once its numerical flattening is repaired, matches the information based acquisitions [4], the max value entropy search matches the costlier entropy searches [45], and at small budgets a greedy exploitation of the posterior mean does as well as either, including on a computational fluid dynamics case [11]. Chapter 5 finds a different ordering on our problem: the greedy local step is the slowest arm when the process has no prior mean, and pays only once a prior gives the model a shape beyond the anchors, an interaction Section 5.5 measures.

Two ingredients matter later. First, the prior mean m need not be zero. A cheap model can supply it, and the process then learns only the residual between that model and the truth. This is the classical idea behind cokriging [24], hierarchical kriging [20], and trust region model management [28, 6]. That such multi fidelity schemes cut the high fidelity cost in aerodynamics is well established [46, 38, 31] and is not a contribution we claim. Second, those schemes assume a clean hierarchy, typically a coarse mesh or a cheaper version of the same solver, whose bias with respect to the truth is consistent and smooth. Our zoo breaks this assumption. Its members have unknown and non uniform biases against RANS, they come from different data and different physics, and there is no shared fidelity ladder. A single correction of the cokriging type is then ill posed, and fusing several members naively injects their errors into the process, a failure we observed and return to in Chapter 4. This is why each member is debiased separately against the RANS reference before it shapes the prior, and whether that correction helps is itself regime dependent, which Chapter 5 measures,

$$m(x) = \sum_m w_m \big(f_m(x) + \hat{\delta}_m(x)\big), \qquad \sum_m w_m \le 1,$$

with f_m the prediction of member m, $\hat{\delta}_m$ its residual model of the next section, and the weight left over given to no prior at all. Chapter 5 sets the weights.

2.5 IO-CUE, uncertainty for a black box against a chosen reference

Uncertainty methods split into two families. White box methods need the inside of a model, its weights, its gradients, or the freedom to retrain it. Deep ensembles, Monte Carlo dropout, and Laplace approximations belong here [27, 17, 10]. Black box methods use only inputs and outputs. A pool of solvers and pretrained networks allows only the second family. There is a deeper reason to prefer it. A white box uncertainty answers the question of how sure a model is about reproducing its own training data. An ensemble of NeuralFoil networks measures disagreement about XFOIL, because XFOIL produced its training data, and it says nothing about the distance to RANS. What the optimizer needs is uncertainty with respect to the reference it will be verified against. Measuring every member on the same held out RANS probe, in black box mode, gives uncertainty with respect to a RANS reference, whatever the origin of the member, and that is what we fit.

It is not the same solver as the one that defines truth inside the loop, and the distinction deserves to be made here rather than discovered later. The probe carries labels from four solver families and none of them is ours. That is deliberate. The largest source of Table 2.1 supplies more than three quarters of the rows with a single solver, and adopting that solver as the reference would have judged the networks of Section 2.1 against the very code that produced their training labels, which would turn the central measurement of Chapter 4 into a test of self consistency. We chose a reference that appears nowhere in the training data, and we pay for that independence in the correction term rather than in the result. Two consequences follow, and we measure both rather than assume them away. The bias a member is credited with is its bias against that family of solvers, and Section 6.3 reports that it differs from the bias against our own solver by about four counts of drag for the global network and sixteen for NeuralFoil, against biases of order two and fifty counts, so the sign of the correction never changes and only its size does. And the conformal guarantee is a guarantee on the probe distribution; whether it survives at the designs an optimizer delivers, measured against our own solver, is the question Section 4.3 answers directly.

We follow IO-CUE, input output conditioned uncertainty estimation [8], in the black box setting of [9]. For a member with prediction $\hat{y}(x)$ the quantity to model is the residual against the reference, $r = f(x) - \hat{y}(x)$, as a function of the pair $u = (x, \hat{y})$. Conditioning on the output as well as the input is what lets the model learn, for instance, that a large predicted lift near stall is suspicious. Instead of a Gaussian head we fit quantile heads with gradient boosted trees [16, 30], chosen over a neural head because the features are tabular, the fit is quick to redo when the probe changes, and a quantile loss does not collapse the spread the way a maximum likelihood Gaussian head can. We then calibrate them by conformalized quantile regression [37, 5, 44] on the calibration part of the probe,

$$E_i = \max\big(\hat{q}_{\mathrm{lo}}(u_i) - r_i,\; r_i - \hat{q}_{\mathrm{hi}}(u_i)\big), \qquad \hat{C}_{1-\alpha}(u) = \big[\hat{y} + \hat{q}_{\mathrm{lo}}(u) - Q_{1-\alpha}(E),\; \hat{y} + \hat{q}_{\mathrm{hi}}(u) + Q_{1-\alpha}(E)\big],$$

where $Q_{1-\alpha}(E)$ is the empirical $(1-\alpha)$ quantile of the scores. Two quantities are read from the fit. The half width of the interval is the uncertainty $\hat{\sigma}(u)$, and the median residual $\hat{\delta}(u) = \hat{q}_{1/2}(u)$ is the debias term used in the prior mean of the previous section.

Figure 2.3 shows why the quantile head is not a detail. The residual of our global network against RANS is sharply peaked with heavy tails, an excess kurtosis near twenty six, and a single Gaussian width cannot fit both the peak and the tails. A Gaussian error bar built on the same conditional scale then covers too much in the bulk and too little in the tail, and the tail is the part a gate would read: at the ninety per cent rate we deploy it reaches eighty six per cent on lift and eighty two on drag. The conformalised conditional quantile tracks the requested rate within one point on both. IO-CUE is thus calibrated in distribution on held out airfoils, and that calibration is checked in every regime of Table 2.2. The guarantee is, however, marginal and not conditional [18]. The interval holds on average over the probe, not necessarily at any particular design. Chapter 4 shows that the designs an optimizer delivers are exactly where this distinction bites.

What the correction does to each member. Figures 2.4 and 2.5 read the zoo against our own solver, on airfoils that neither the members nor the correction ever saw, before and after $\hat{\delta}$. Competence

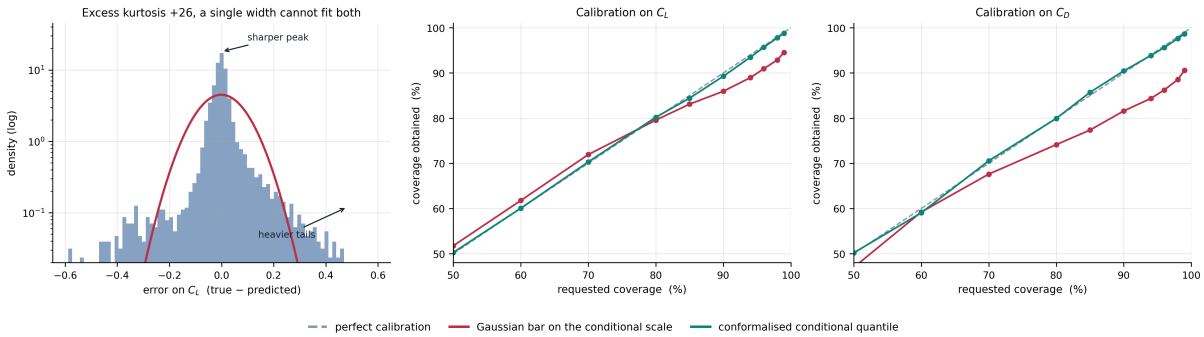


Figure 2.3: IO-CUE on the global network, held out probe. *Left:* the residual against RANS on a logarithmic density axis, with a Gaussian of the same variance, showing the sharper peak and the heavier tails. *Middle and right:* coverage obtained against coverage requested, on lift and on drag, for the deployed heads. Both are conditioned on input and output and fitted on the prediction split; the Gaussian bar scales the conditional mean absolute residual, the conformal interval adds an offset fitted on one half of the calibration split. Coverage is measured on the other half, so it is held out in both cases.

has two axes, ordering designs and estimating their value, and the ranking of the members changes with the axis one reads. Raw, four of the seven members order the attached designs with a Kendall rank correlation of +0.80 or above, and all seven order the transonic ones at random, between −0.10 and +0.01. The correction leaves the global network almost unchanged, since it was trained on every source the correction was fitted on, and it transforms the members trained on a single source or on another fidelity: HAM2D goes from +0.06 to +0.42 near stall, AirfoilANN from +0.06 to +0.66 in attached flow, NeuralFoil from −0.02 to +0.74 in transonic flow. The correction is worth the bias it has to remove.

It also has a limit, which the polars make visible. At Mach 0.734 and 0.80 the true drag rises steeply with incidence as the shock forms. The global network and the subsonic compressible expert stay flat near ninety counts: they do not represent the shock, and a correction fitted on residuals moves their level without giving them the rise. In transonic flow their order stays near random after correction and their drag error near eighty per cent. NeuralFoil and the attached expert do have a rise, and the correction restores their order, but on the RAE case the rise of the corrected NeuralFoil comes too early, and at three degrees it announces two to three times the true drag. The correction cannot supply physics a member lacks, and where the member has it, it may place it wrongly. The stall of the NACA 0012 at Mach 0.15, near seventeen degrees, is seen by no member.

What the toolbox does not yet say. Nothing above guarantees that the uncertainty of Section 2.5 stays informative at the designs an optimizer will propose, nor that a gate built on it saves a single RANS solve. Those are the questions of the next chapter, which states the problem and the hypotheses that Chapter 4 then tests.

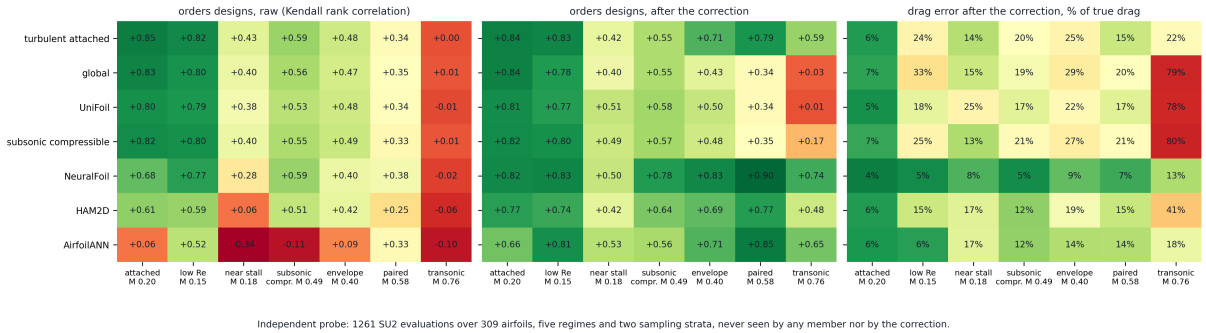


Figure 2.4: Competence of the zoo regime by regime, judged against our own solver on an independent sweep of 1261 evaluations over 309 airfoils that neither the members nor the residual correction ever saw. Two columns are sampling strata of the probe rather than flow regimes: *envelope*, single points drawn uniformly over the flight envelope, and *paired*, conditions at which an open source also provides a RANS value. *Left:* how well each member orders designs by lift to drag, raw. *Middle:* the same after the correction. *Right:* drag error after the correction, as a share of the true drag of that regime.

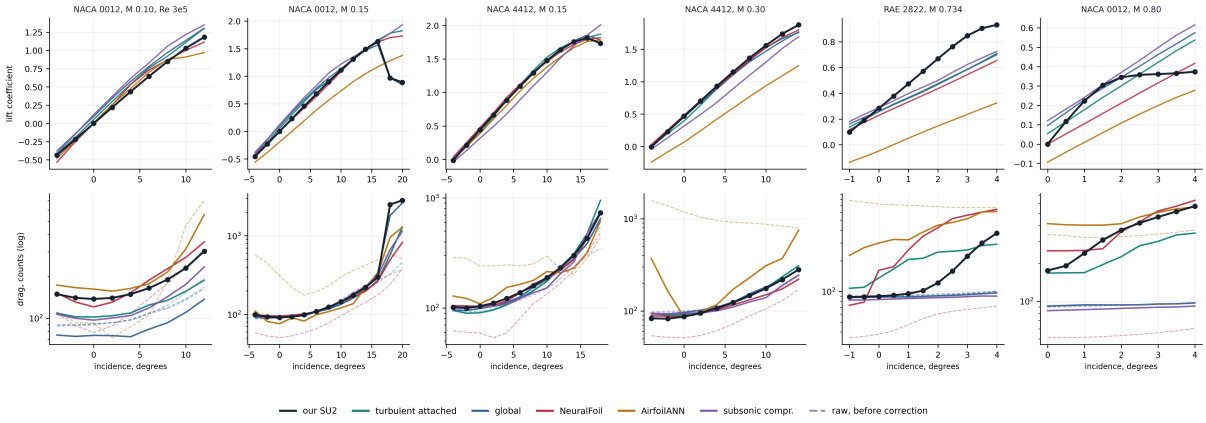


Figure 2.5: The same members read as polars, on six sweeps run with our solver. *Top:* lift against incidence. *Bottom:* drag against incidence on a logarithmic axis, solid after the residual correction and faint dashed before it. The gap between the two is what the correction buys, and it is widest where the member is furthest from the data it was trained on. Five of the seven members of the pool are drawn, the two source experts are left out for legibility.

Chapter 3

Problem and hypotheses

This chapter states the problem in one page, the single property the uncertainty must satisfy for the bet of the introduction to work, the three hypotheses the thesis tests, and the protocol and the cases on which they are tested.

3.1 The optimization problem

We seek the shape of a two dimensional airfoil that optimizes an aerodynamic performance at given flight conditions, with a fixed number of high fidelity evaluations. The shape is the twenty two parameter CST vector of Section 2.3, searched through its eight POD coordinates, a choice Chapter 5 shows accelerates the optimizer. The performance is computed by the RANS solver that serves as reference. The budget is one hundred RANS evaluations per run, all included, the starting design, the initialisation, the loop, and the final verification.

Two objectives are used. The first maximises the lift to drag ratio $L/D = C_L/C_D$ at fixed incidence, the natural continuation of the simple use case. The second minimises drag, at a prescribed lift, the classical formulation of design benchmarks [29], or at fixed incidence for the transonic tutorial case. They are two among a wider family used in industry, weighted drag over several cruise points, off design robustness, stall margin, or structural thickness constraints, and we take the two canonical single point formulations, leaving the multi point case to future work. Both are subject to a design box centred on the starting airfoil and to manufacturability rules that are geometric and fixed from the literature, not from our data. A candidate is admitted only if its curvature changes sign at most twice over the two surfaces, which excludes wavy shapes [25], and if its thickness at two per cent of the chord is at least two per cent of the chord, which excludes a knife edge leading edge [1]. A verified design must in addition be no thinner than the starting airfoil, and at prescribed lift must hold the target lift within 0.02,

$$\text{(i)}\quad \max_{x\in\mathscr{X}} \frac{C_L(x)}{C_D(x)} \ \text{ at fixed } \alpha, \qquad\qquad \text{(ii)}\quad \min_{x\in\mathscr{X}} C_D(x) \ \text{ at } C_L(x) = C_L^\star \ \text{ or at fixed } \alpha,$$

$$\text{both subject to } g_{\text{manuf}}(x) \le 0 \ \text{ and } \ \#\{\text{RANS calls}\} \le B = 100.$$

One design decision matters for everything that follows. The starting airfoils are deliberately far from optimal for their conditions, a symmetric profile at positive incidence that must grow camber, or a transonic airfoil placed off its design point, so that the achievable gain is large. The economy of a method is then measured over a real descent of the objective, not inside the noise between seeds.

3.2 The property uncertainty must satisfy

To accept a surrogate value in place of a RANS solve, the error of the surrogate with respect to RANS must be small at that design. That error is unknown at decision time, so a gate reads the uncertainty

$\hat{\sigma}$ as its proxy and trusts the surrogate whenever $\hat{\sigma}$ falls below a tolerance. The property required is therefore that a small uncertainty implies a small error to the reference, at the designs the optimizer selects and not on average over a test set,

$$\text{gate:} \quad \text{trust } \hat{y}(x) \text{ iff } \hat{\sigma}(x) < \tau,$$

$$\text{required:} \quad \mathbb{P}\big(|f(x^\star) - \hat{y}(x^\star)| \le \hat{\sigma}(x^\star)\big) \approx 1-\alpha \quad \text{and} \quad \rho\big(\hat{\sigma}, |f-\hat{y}|\big) > 0 \text{ over delivered designs } x^\star.$$

This is not guaranteed by what Chapter 2 established. The calibration of IO-CUE is marginal, it holds on average over the probe. The designs an optimizer selects are by construction those where the surrogate is most optimistic, and nothing so far says the property holds there.

Two warnings about this property, because the result of Chapter 4 turns on both. It is stated at the delivered designs and not on a test set, which is the whole point, and a calibration that holds on average over a population says nothing about a subpopulation chosen by taking a maximum. And it is a necessary condition, not a sufficient one. Chapter 4 finds that both halves of it hold in the constrained search, the nominal band covers what it should and the uncertainty ranks the error, and that the gate still saves nothing. The reason is not that the uncertainty is wrong but that trust is granted only where it is worthless, which no amount of calibration repairs. The reader should therefore not expect the refutation to take the form of a violated inequality.

Two uses of the uncertainty rest on this property and are evaluated separately. As a gate, deciding whether to trust or to verify a point. As a selector, choosing at each point the member of the zoo with the smallest uncertainty.

One consequence of the gate rule must be stated before any result. If a surrogate is technically good on a case, its uncertainty is small there and the gate should trust it often. If no member is trusted, the router verifies every point and is the reference loop under another name, and no economy can appear. Economy is therefore expected only where trust actually happens, and a test of the gate is meaningful only on cases where one member of the zoo is trusted for a substantial share of the search.

3.3 Hypotheses

H1 A gate on $\hat{\sigma}$ reaches the same verified performance with fewer RANS evaluations than verifying every proposal. *(Tested in Section 4.2.)*

H2 The error of the surrogate at the delivered design is detectable by its uncertainty. *(Tested in Section 4.3.)*

H3 Inside the gated loop, selecting the zoo member point by point by its uncertainty improves on using a single member. *(Tested in Section 4.2.)*

Chapter 4 tests H1, H2 and H3. Chapter 5 shows where the economy comes from when it exists, and how the prior is chosen.

3.4 How we measure

All measurements are taken on the designs the optimizer actually proposes, never on a random probe, because that is where selection bias acts. The truth is always a RANS evaluation of the design in question. Every configuration is repeated on three seeds, 42, 123, and 456, and the spread between seeds is reported with every curve.

Two search spaces appear in the thesis. The constrained space, the POD latent with eight modes, a narrow box around the starting airfoil and the manufacturability rules, is the default and the regime in which designs are delivered. The free space, the twenty two CST coefficients in a wide box without manufacturability rules, appears only where the contrast is the point, the curse contrast of Chapter 4.

Fidelity. The RANS evaluations inside the loop use a fast setting of the solver, the same mesh with fewer iterations, and the delivered design of every run is re evaluated at full convergence. Figure A.5 in the appendix compares the two on the delivered designs of the reference loop. Twenty seven of the twenty eight delivered designs agree within 1.2 units. The exception is one transonic run, case 4, the off design RAE case, on one seed, where the fast setting under resolves the shock and announces 80.7 where full convergence gives 57.4. Every curve of this thesis is read at the fast setting, the one every arm used, so comparisons between arms are fair; the absolute values of the transonic cases carry that reserve. Runs that are preempted on the cluster resume from their last verified design with all their anchors and their remaining budget.

The method is validated, or not, on nine cases, Table 3.1, chosen by six principles. The same starting airfoil across most cases, so that the regime and not the shape is what changes. Both objectives. Every expert of the zoo exercised at least once. At least one case where NeuralFoil is blind by construction, the transonic shock. At least one cambered geometry at another Reynolds number. Starting designs off their design point, so that there is room to improve.

Table 3.1: The nine optimization cases. Objective (i) is maximum lift to drag at fixed incidence, (ii) is minimum drag, at prescribed lift or at fixed incidence. Regime labels as in Table 2.1. Figures name a case by its number, its objective and its Mach number, written M.

#	case	start	Mach	Re	α or $C_L^\star$	obj.	regime	what it tests
1	Max L/D	NACA 0012	0.15	6×10^6	$\alpha = 5°$	(i)	TA	anchor, camber must emerge
2	Max L/D	NACA 0012	0.50	6×10^6	$\alpha = 3°$	(i)	SC	intermediate regime, no shock
3	Max L/D	NACA 0012	0.30	6×10^6	$\alpha = 4°$	(i)	TA	light compressibility
4	RAE 2822	RAE 2822	0.734	6.5×10^6	$\alpha = 3.0°$	(i)	TR	off design, shock, NeuralFoil blind
5	Low Re	NACA 0012	0.10	3×10^5	$\alpha = 5°$	(i)	LR	laminar bubble, transition
6	Near stall	NACA 0012	0.15	6×10^6	$\alpha = 9°$	(i)	NS	separation, experts unreliable
7	NACA 4412	NACA 4412	0.30	1.2×10^7	$\alpha = 3°$	(i)	TA	other geometry, other Re
8	Min CD	NACA 0012	0.15	6×10^6	$C_L^\star = 0.5$	(ii)	TA	fixed lift, well posed
9	Min CD	NACA 0012	0.80	3×10^6	$\alpha = 1.25°$	(ii)	TR	SU2 tutorial, strong shock

The validation criterion is the same for every case. Take the verified performance reached by the reference optimizer, without prior, after its one hundred RANS evaluations. A method is validated on a case if its verified performance reaches ninety eight per cent of that reference, and we report the number of RANS evaluations it needed to get there. For drag the criterion is symmetric, staying within two per cent above the reference drag. A case where the threshold is never reached within the budget counts as a failure, not as a near miss. The threshold is set at ninety eight per cent because a coarser one would validate methods that differ only by noise. Two per cent is of the order of the spread between seeds of the reference loop on four cases, cases 2, 6, 7 and 8, from 0.3 to 2.0 per cent. On four others, cases 1, 3, 5 and 9, the reference itself spreads by 2.4 to 10 per cent between seeds, the threshold falls inside that noise, and there we report how many of the three seeds cross it rather than a single crossing.

Chapter 5 applies this criterion case by case, and Figure 5.10 gathers the verdict on the nine cases.

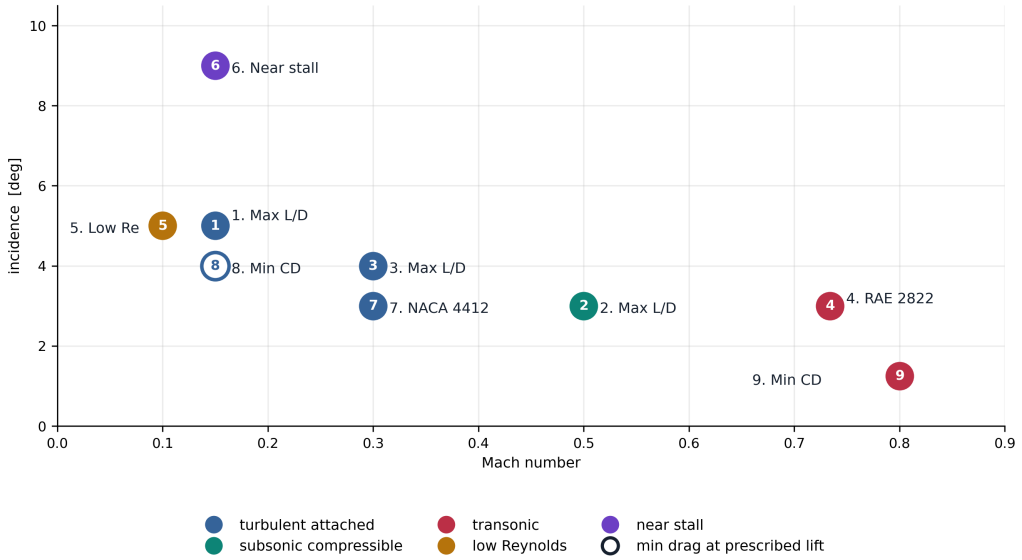


Figure 3.1: The nine test cases in the Mach against incidence plane, coloured by flow regime and numbered as in Table 3.1. The hollow marker is the prescribed lift case, placed at its initial incidence. The cases cover five regimes and not only the comfortable attached zone.

3.5 The optimizer we test

Every result in this thesis comes out of one Bayesian optimization loop, and its variants differ by one mechanism at a time. The loop searches the normalised design space, the eight POD coordinates mapped to the unit cube, and it learns the truth only through RANS evaluations counted in the budget. Two configurations of this loop serve as baselines and they play different roles. The *reference loop*, arm B, uses the eight mode latent space, the PIMS acquisition, the probability that a candidate improves on the maximum of one posterior sample path [43], defined with four others in Section 5.3, no prior mean, and verifies every proposal with RANS. It is the reference of the validation criterion of Section 3.4 and the arm the router is compared to in Chapter 4. The *first baseline*, arm A, uses expected improvement (EI) in the raw twenty two coefficient space, no prior mean, and verifies every proposal. It is the historical starting point, and Chapter 5 measures the path from it to the reference loop and beyond, through an intermediate arm A′, expected improvement in the eight mode latent space, which isolates the latent lever.

Algorithm 1 gives the loop in nine steps, and the paragraphs below detail each step.

Algorithm 1. The optimization loop. In the margin, the mechanism each chapter varies.

1. Evaluate the starting airfoil with RANS, set y_{base}.
2. Draw five manufacturable Sobol points in the box, evaluate them with RANS.
3. Fit the Gaussian process with per point noise (prior mean and online prior weights, Chapter 5)
4. Draw two thousand manufacturable candidates. (latent space, trust region, Chapter 5)
5. Pick the candidate by the acquisition. (PIMS, or a trust region with a prior, Chapter 5)
6. Decide to trust or to verify the candidate. (gate and selector, Chapter 4)
7. Verified, add the RANS value with floor variance, update the best verified design. Trusted, add the belief with its squared uncertainty.
8. Repeat from step 3 until the RANS budget is spent.
9. Verify the K best unverified beliefs with RANS, deliver the best verified design.

Chapter 4 runs this loop as written. The arms of Chapter 5 are special cases of it: their tolerance is zero, so step 6 always verifies and step 9 has nothing left to verify, and in the arms with a prior, steps 4 and 5 are replaced by the trust region step described below.

Initialisation. The starting airfoil is evaluated with RANS, which sets the base performance y_{base} that scales every tolerance. Five further points are then drawn by a Sobol sequence in the design box, redrawn until five manufacturable shapes are obtained, and each is evaluated with RANS, six RANS evaluations in total with the starting design. These anchors seed the Gaussian process.

A Gaussian process with per point noise. The model is a Gaussian process on the standardised objective with an observation noise variance that is specific to each point. A RANS evaluation enters with a floor variance v_0, the solver being nearly deterministic. A surrogate belief accepted without verification, when a gate allows it, enters with the squared uncertainty of that belief. With n observations y at points X and noise variances v,

$$\mu_n(x) = m(x) + k(x,X)\left(K + \operatorname{diag} v\right)^{-1}\left(y - m(X)\right), \qquad s_n^2(x) = k(x,x) - k(x,X)\left(K + \operatorname{diag} v\right)^{-1} k(X,x),$$

$$v_i = v_0 \text{ for a verified point}, \qquad v_i = \hat{\sigma}_i^2 \text{ for a trusted point.}$$

This bounds the weight of an uncertain point on the posterior mean. The kernel is a squared exponential kernel with one length scale per dimension, under the log normal length scale prior scaled with the dimension that recent high dimensional practice recommends [21]. The prior mean m is zero in the reference loop and becomes the debiased surrogate in the variants of Chapter 5.

Candidates and acquisition. The acquisition proposes one design per iteration, chosen inside a pool rather than by a continuous optimization: two thousand candidates drawn by a Sobol sequence in the box, filtered by the manufacturability rules, redrawn with a fresh seed at every iteration. The pool is not a shortcut. The manufacturability filter is not differentiable, so the acquisition cannot be maximised by gradient under it, and in eight dimensions a dense pool approximates the continuous maximum closely. The acquisition of the reference loop is PIMS, defined with the four others in Section 5.3; what matters here is that it has no parameter to tune and that its sample path is seeded by the iteration index, so two arms that have seen the same data see the same acquisition. With a prior mean, Chapter 5 replaces this step by a trust region step, the maximum of the model inside a region around the current best, the region resized by the ratio of real to predicted improvement and restarted when it collapses.

Verification and stopping. In the reference loop every selected candidate is evaluated with RANS and enters the process with the floor variance, and the best verified design is updated. The loop stops when the RANS budget is spent. A patience on the plateau of the best verified design exists and is switched off in this thesis. A trust region without restart stops when the region contracts, after a few tens of RANS evaluations, so the arms that use a trust region, Chapter 5, restart it when it collapses and run to the full budget like the others.

Final verification. At the end, the K most promising designs according to their belief that have not been verified are evaluated with RANS, within the total budget. The delivered design is always the best verified one. This rule is the same for every arm, and no performance is reported in this thesis without a RANS evaluation behind it.

Chapter 4

The uncertainty gate does not hold

4.1 The chosen design disappoints when verified

When a design is chosen by maximising an estimate, the realised value of the winner is on average below its estimate, even when the estimator is unbiased on average. This is the post decision surprise of Smith and Winkler [41], the optimizer's curse,

$$\mathbb{E}\big[\,\hat{f}(x^\star) - f(x^\star)\,\big] > 0, \qquad x^\star = \arg\max_x \hat{f}(x).$$

In our loop the estimate is the debiased belief of the surrogate and the winner is the design retained. The phenomenon is sharpened by the nature of the problem. The surrogate learned on realistic airfoils and the optimizer looks for extremes, so it retains the candidates on which the surrogate is wrong in the direction it likes.

Figure 4.1 shows it on the archive of our constrained runs, with one unit throughout. The left panel gathers every verified design. The cloud lies on both sides of the no bias line: over the population the surrogate errs both ways. The believed optimum of each run, in red, lies almost always above it. The designs of the transonic runs sit apart, below the line, because that is where the member under predicts hardest, the far side of the same error. The middle panel shows the same surrogate on the held out airfoils of the probe, in the regimes of cases 1 to 3: its error is centred on zero. The right panel shows it at the believed optimum of each run: about ten units too optimistic in median. Nothing changed in the model between the panels, only the way the points were chosen. This surrogate is not a poor model; this is the signature of selection, an error symmetric over the population and positive on the elected.

Had the loop delivered its believed optimum without verifying it, the right panel is what it would have delivered: a phantom, a design whose belief exceeds its RANS value, in all but a handful of runs. The final verification is what catches this, at the price of one RANS evaluation per delivered design, and Section 4.4 draws the consequence. What remains to be asked, and is the object of the next section, is whether the verification during the search can be economised by a gate on the uncertainty.

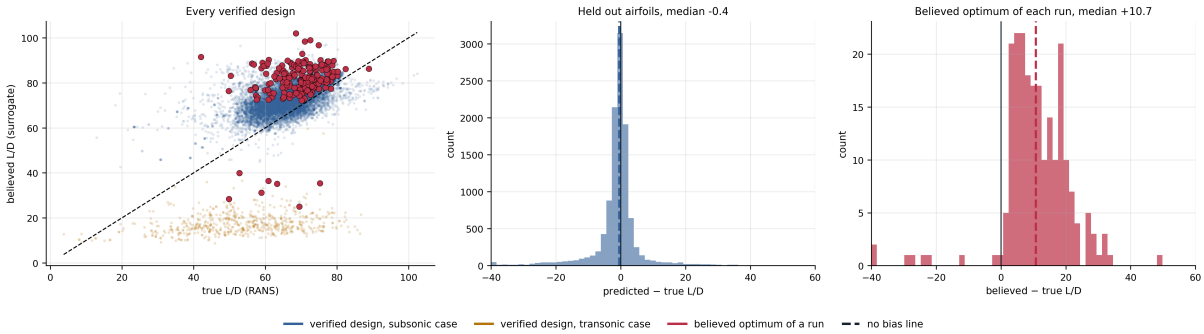


Figure 4.1: The curse is a selection effect, not a poor surrogate. Left, believed against true lift to drag ratio at every verified design of the constrained runs of our archive, subsonic and transonic cases drawn apart, the believed optimum of each run in red. Middle, the error of the same surrogate on the held out airfoils of the probe, attached and compressible regimes. Right, the optimism gap at the believed optimum of each run, on the same axis as the middle panel: what the loop would hand over without a final verification.

4.2 Changing the gate policy does not help

The router is the reference loop with two additions at step 6 of Algorithm 1, a selector and a gate.

Selection by uncertainty. For a candidate x, every member m of the zoo returns a prediction $\hat{y}_m(x)$, and the uncertainty model of Section 2.5 returns $\hat{\sigma}_m(x)$ from the inputs and that prediction. The router keeps the member with the smallest uncertainty,

$$m^\star(x) = \arg\min_m \hat{\sigma}_m(x), \qquad \hat{y}^\star(x) = \hat{y}_{m^\star}(x), \qquad \hat{\sigma}^\star(x) = \hat{\sigma}_{m^\star}(x).$$

This is model selection point by point by uncertainty, hypothesis H3. Two variants are tested as well, a fusion of the members weighted by inverse variance, and a choice by the real error measured at the nearest RANS anchors.

The gate. The tolerance is set relative to the base performance, $\tau = 0.2\,|y_{\text{base}}|$ in the configuration retained, and the rule is

$$\hat{\sigma}^\star(x) < \tau \implies \text{trust}, \qquad \text{otherwise} \implies \text{verify}.$$

A trusted point enters the Gaussian process with its belief $\hat{y}^\star$ and the variance $\hat{\sigma}^{\star 2}$, costs no RANS evaluation, and never updates the best verified design. A verified point enters with the RANS value and the floor variance. This is hypothesis H1, the gate should reach the same best verified design with fewer RANS evaluations. Several gate policies were tried, a fixed tolerance, a tolerance that shrinks with the budget spent, a local margin relative to the current best, and an escalation to an intermediate fidelity. None changes the conclusion below.

What is compared. The full RANS arm and the router arm are identical in everything else, the same initialisation, the same process, the same acquisition with the same seeds, the same candidates at equal data. Only the verification rule differs, so any difference between the curves is due to the gate and the selector, and to nothing else. Following Section 3.2, the share of proposals trusted is reported with every router curve, and hypothesis H1 is judged on the cases where that share is substantial. The zoo and the selector are there to make trust likely, on any given regime some member usually carries a small uncertainty, which is what should produce trust and economy if the property of Section 3.2 held.

Reading the curves. Figure 4.2 shows the nine cases of Table 3.1, the full RANS arm and the gated router, median over three seeds with the band between seeds, and in each panel the share of loop proposals the gate trusted. Three groups appear. On two cases the gate never or almost never trusts, the low Reynolds case and the transonic minimum drag case, and the two arms coincide exactly. This is the check announced in Section 3.5, with the gate closed the zoo never enters the Gaussian process and the router is the reference loop. On five cases the gate trusts between one proposal in fifteen and one in two, and the router curve follows the full curve for the first half of the budget, then flattens below it. On the two remaining cases the gate trusts almost everything, the near stall case and the incompressible minimum drag case. There the loop is paced by the surrogate and not by RANS, hundreds of proposals pass for every verification, and both were run with the iteration cap of the harness lifted, so that the RANS budget is what stops them. On the near stall case the router ends below the full arm. On the incompressible minimum drag case it reaches the target with the same number of evaluations and ends level with it. On no case does the router reach the plateau of the full arm earlier, which is what hypothesis H1 requires.

Why the router ends below. A trusted point enters the Gaussian process with a belief and never becomes a verified design. Two consequences follow. The trusted beliefs pull the process toward the regions the surrogate likes, so the next proposals concentrate there, and when that region is a phantom in the sense of Section 4.1 the RANS evaluations that follow are spent confirming a disappointment. And a trusted proposal is verified progress deferred, not gained, the best verified design only moves when RANS is spent, so a gate that trusts nearly everything makes the loop long without making it better. The selector does not repair this. Figure A.2 in the appendix shows the member picked at each proposal on case 1, the expert trained on attached turbulent flow dominates and the choice is stable across the run. The difficulty is not which member is believed, it is that believing any member at the proposals costs verified progress.

The tolerance has no sweet spot. Figure 4.3 sweeps the tolerance on case 1, on one seed of an earlier campaign whose level sits above the median of Figure 4.2; the sweep is read for its trend, not its level. As τ grows the share trusted grows from nothing to nearly everything. The verified best does not move for small tolerances, the trusted points are few and far from the optimum, and degrades for large ones, the loop starts believing phantoms and the largest trusted belief above the true optimum grows with τ. Between the two there is no tolerance at which the router saves RANS without losing design. This is the dose response expected when the property of Section 3.2 does not hold, trusting more only buys more phantoms.

Other policies, trust weights and selectors. The fixed tolerance is not the only gate we tried. Figure A.7 in the appendix gathers, on case 1 and one seed, eight gate policies, a fixed tolerance, a tolerance shrinking with the budget, a local margin to the current best, a hybrid of the two, three escalation rules and a convex shrink, four ways of weighting a trusted point in the Gaussian process, and three selectors, the uncertainty argmin over a curated zoo, an inverse variance fusion of the members, and a choice by the skill measured at the RANS anchors. Every variant that trusts little lands within a point or two of the full arm, which is the spread between seeds. Every variant that trusts a lot ends below it. None rises above it by more than the spread between seeds. The policies differ in when they trust, not in what a trusted point does to the loop, and that is where the loss sits.

Controls. Two controls isolate the mechanism. The first is a gate that trusts at a comparable rate, but on proposals drawn at random. On case 1 it ends at 79.5 in median, between the uncertainty gate, 77.1, and the full arm, 80.3. Choosing which proposals to trust by their uncertainty buys nothing over chance, and here does slightly worse. The second keeps the gate and removes the selection: the router may believe only the global member. On case 1 it reaches the target in sixty eight evaluations where the zoo router never does; on cases 2 and 3 neither reaches it, and the single member ends level with

the zoo on case 2 and one and a half units below on case 3. Selecting the member point by point adds nothing systematic to the best single member.

Verdict. Hypothesis H1 is refuted on the nine cases, and hypothesis H3 on the three where it was tested. The gate is not miscalibrated, where it trusts little it is right to, and where it trusts a lot the design suffers. It does not reduce the number of RANS evaluations needed, and the next section says why.

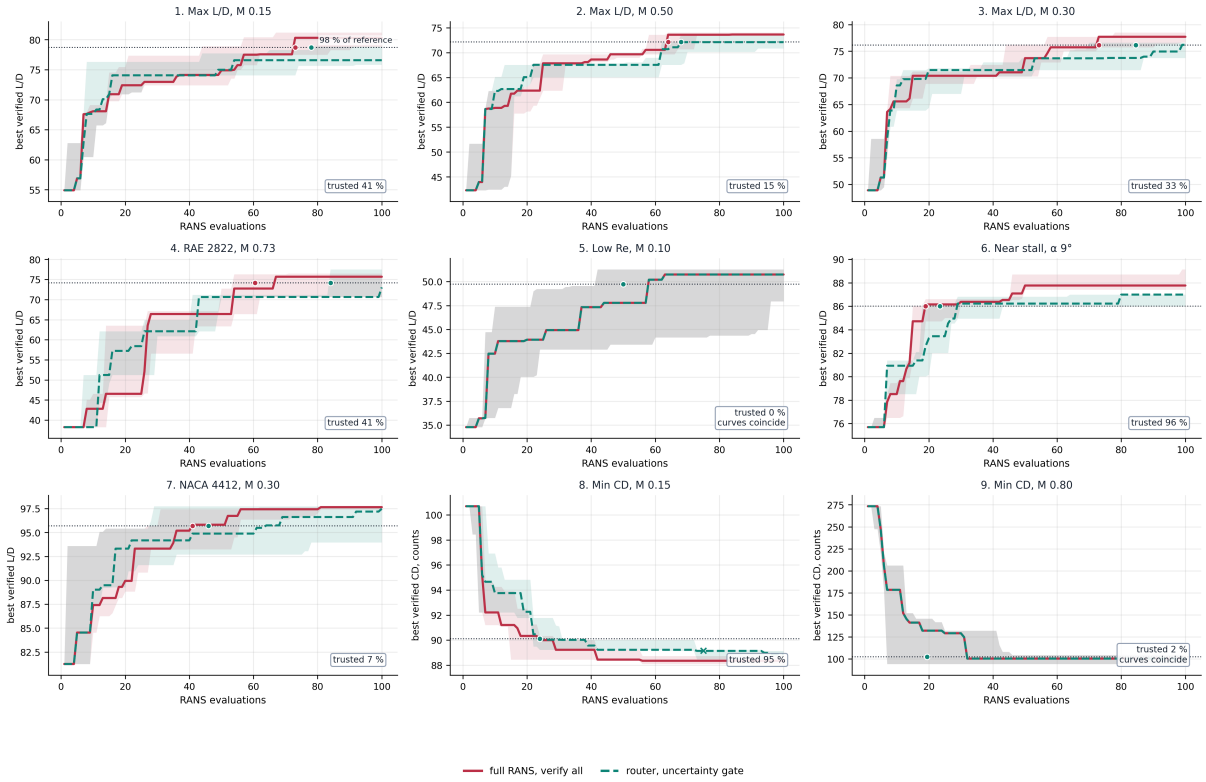


Figure 4.2: Verified best design against RANS evaluations spent on the nine cases of Table 3.1, full RANS arm in red and gated router in green, median of three seeds with the band between seeds. The dotted line is ninety eight per cent of the plateau of the full arm, the reference loop, and a dot marks where a curve crosses it. Each panel gives the share of loop proposals the gate trusted. Where the gate never trusts the two curves coincide, which confirms that the full arm is the reference loop without a gate. Where it trusts some, the router follows and then flattens below. A cross marks where the first seed of an arm ended, after which the curve holds its best verified design.

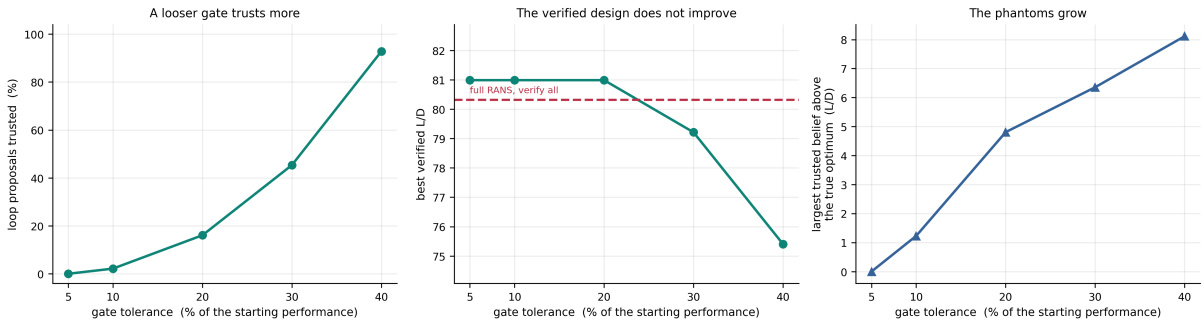


Figure 4.3: Tolerance sweep of the gate on case 1, one seed of an earlier campaign. Left, the share of proposals trusted grows with the tolerance. Middle, the verified best design does not improve and eventually degrades below the full RANS arm. Right, the largest trusted belief above the true optimum grows. The three quantities have their own axis and are not comparable to one another.

4.3 Why the gate does not economize, the designs worth trusting are the uncertain ones

To understand why the gate does not economize, the error must be measured where the optimizer chooses, and compared with what the uncertainty announced there. We therefore run a controlled experiment in which the loop trusts every proposal during the search and verifies with RANS its twenty five best believed designs at the end. Each selected design yields a triplet, the belief of the surrogate, its uncertainty, and the RANS truth. Cases 1 and 2, three seeds, and two search spaces are used, the constrained space of the rest of this thesis, and a free space in raw CST coefficients, wide, without manufacturability rules, to see what happens when the optimizer is allowed to leave the data. Figure 4.4 gathers the result.

In the constrained space the selected designs stay as close to the training data as the reference airfoils themselves, right panel. Their optimism is real, a median gap of about sixteen units between belief and truth, left panel, more than the ten of Figure 4.1 because these runs trust every proposal, so nothing corrects the surrogate during the search. And the uncertainty does its job. It ranks the designs by their error, with a rank correlation near one half, and its magnitude is right, the nominal ninety per cent band covers ninety two per cent of the realised errors, middle panel. In this space the uncertainty model of Section 2.5 keeps its promise all the way to the delivered design. Figure A.1 in Appendix A shows the same behaviour on the whole archive of earlier constrained runs.

Left free, the search leaves the data. The selected designs sit twice as far from the reference airfoils, and the optimism explodes to a median of sixty. One design in five could not even be evaluated, the geometry does not mesh or does not converge. The uncertainty still ranks the error, with the same correlation, but it no longer measures it, the nominal band covers two thirds of the cases. This is the conditional coverage failure announced in Section 2.5, a calibration guaranteed on average over the data says nothing outside the data. The two searches differ in three ways at once, the manufacturability rules, the eight mode latent space instead of the twenty two coefficients, and a box three times narrower, and we do not separate them. Together they are not a detail: they bound the curse from sixty to sixteen and restore the coverage, which Section 4.4 takes up.

It remains to explain the result of Section 4.2 in the constrained space, where the uncertainty is right and the gate still saves no evaluation. The mechanism is a selection effect. Among candidates of comparable true value, the optimizer retains those whose residual happens to be positive, so the residual of the selected designs is positive even when the residual of the surrogate is unbiased on average, which the debiasing of Section 2.5 aims at on the probe set. This is the post decision surprise of Smith and Winkler [41], and it is visible in the numbers, the median gap at the selected designs is of the order of the median uncertainty there, which is what selecting the maximum of estimates with that spread produces. The consequence for the gate follows. At the selected designs, error and

uncertainty are large together, above the tolerance of the gate, a fifth of the starting performance, drawn as the vertical line in the middle panel, so the gate refuses them and verifies, rightly. When it trusts, the uncertainty is small, the candidate lies in a region already anchored by RANS evaluations, and its belief adds nothing to the Gaussian process. The curves of Figure 4.2 follow each other for this reason, and part ways only where the gate trusts enough to feed the process with phantoms. What the router trusts does not matter, and what would matter it cannot trust. Nor is the tolerance at fault. Figure 4.3 showed that any tolerance below the typical error refuses the promising designs and any tolerance above accepts wrong values. Hypothesis H2 must therefore be read with care. The error at the delivered design is detectable, in rank in both spaces and in magnitude in the constrained space. What fails is the premise of the gate, that a calibrated uncertainty suffices to economize. It does not, because trust is granted only where it is worthless.

For a gate to earn, what would have to change is not the uncertainty but the error of the surrogate at the promising designs. Recalibrating the uncertainty would change nothing, the selection effect is untouched. Three routes act on the error itself. Two are tested in this thesis. Correcting the surrogate locally with the RANS anchors as they are paid for, which is what the Gaussian process on the residual does once the surrogate becomes its prior mean, the pivot of Chapter 5. Replacing the surrogate by a source whose bias is consistent and correctable, a coarse RANS, also in Chapter 5. The third, enriching the surrogate between runs with the optima already verified, is left to the perspectives. In every route the verification remains, which the next section establishes.

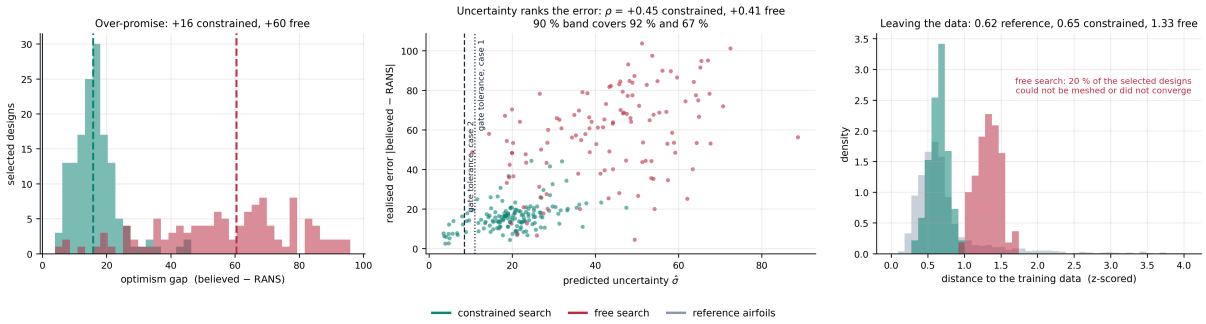


Figure 4.4: Constrained against free search, twenty five selected designs per run, two cases, three seeds. Left, the optimism gap at the selected designs, believed minus RANS, median as a dashed line. Middle, predicted uncertainty against realised error at those designs, with the gate tolerance of each of the two cases drawn as a vertical line. Right, the distance of the selected designs to the training data, against the reference airfoils, with the share of free designs that RANS could not evaluate at all. The constrained search stays on the data and keeps a moderate gap, the free search leaves the data and its gap explodes, while the rank correlation is similar in both.

4.4 Consequence: verification is mandatory, constraints only bound the curse

Verification cannot be skipped. The three previous sections point the same way. The design the optimizer believes best disappoints when verified, Section 4.1, the gate that decides when to skip verification saves no evaluation whatever its policy, Section 4.2, and the reason is a selection effect that no calibration can undo, Section 4.3. The curse is present whenever a maximum is taken over estimates, so it is present in every surrogate assisted optimization, and the only operation that removes it is evaluating the delivered design with the reference. Figure 4.1 showed what happens otherwise, the believed optimum handed over without RANS is optimistic in almost every run. Verifying the delivered design is therefore not a precaution added to the optimizer, it is part of it.

Nor can verification be deferred to the end. One could hope to let the surrogate run the whole search alone and to spend RANS only once, on its favourites. Figure 4.5 tests this on cases 1 and 2. The surrogate guides the loop with no verification at all, then its twenty five best believed designs are verified, which costs the same number of RANS evaluations as the initialisation plus twenty five. The result lands at the level of where the full RANS loop already was after the same number of evaluations, and far below where the full loop ends with its whole budget. The RANS evaluations spent inside the loop do double duty, they verify a design and they teach the Gaussian process where the surrogate is wrong, so that the next proposal moves. Spent at the end they only verify. Economy cannot come from where verification is placed.

Constraints are the cheap lever, and they do not finish the job. Figure 4.4 compared the same search with and without the manufacturability rules and the latent space. The constraints divide the gap at the selected designs by about four and bring the uncertainty back to its nominal coverage. Constraints therefore go in by default, and every arm of Chapter 5 carries them. But in the constrained space the residual gap is still above the tolerance of the gate, and Section 4.2 showed the gate still earns nothing there. Constraints shrink the curse, they do not remove it.

What remains for economy. If verification can be neither skipped nor deferred, economy must come from needing fewer verifications to reach the plateau. Three levers remain, and they are the subject of the next chapter. Search in a space where each RANS evaluation goes further, the latent space. Place the next RANS evaluation better, the acquisition. Start from a mean that is already close to the truth, the prior. In all three the surrogate returns as a guide corrected at every RANS anchor, never as a substitute for one.

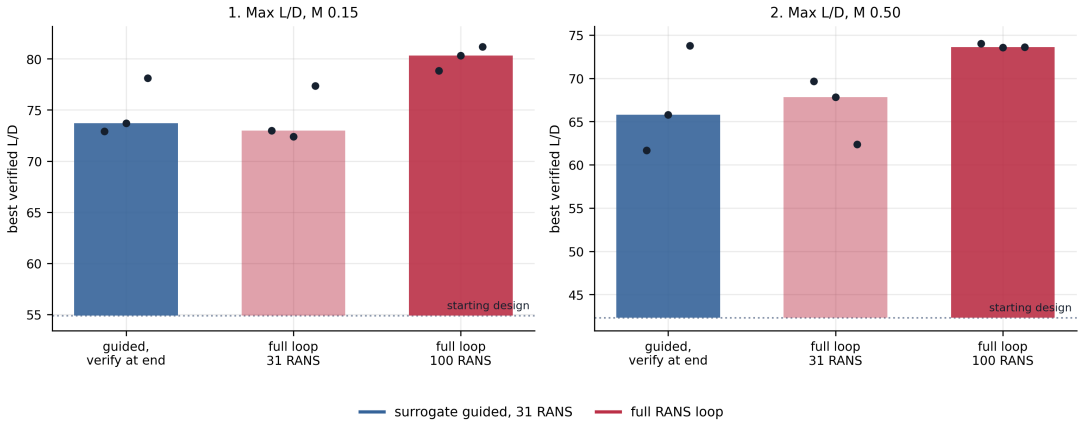


Figure 4.5: Deferring verification to the end, cases 1 and 2. Left bar, the surrogate guides the whole search alone and its twenty five best believed designs are verified at the end, thirty one RANS in total. Middle, the full RANS loop stopped at the same thirty one evaluations. Right, the full RANS loop with its whole budget. Three seeds, dots are the seeds and the bar is the median. Bars start at the starting design, dotted, so their height is the gain actually bought. The deferred verification lands at the level of where the loop already was at equal cost.

Chapter 5

Sources of economy: latent, acquisition, prior

Chapter 4 closed one door. The surrogate cannot replace a RANS evaluation, whatever the gate that decides it. This chapter opens another. The surrogate guides where the RANS evaluations are spent, every design is verified, the budget is fixed, and the question becomes how many evaluations it takes to reach the plateau. Three mechanisms of Algorithm 1 are varied one at a time, the search space, the acquisition and the prior mean, and each is measured against the arm before it on the nine cases of Table 3.1.

5.1 The pivot and the starting baseline

The quantity measured in this chapter is the one defined in Section 3.4, the verified best design against the RANS evaluations spent, and the number of evaluations needed to reach ninety eight per cent of the plateau of the reference loop. The economy of a mechanism is the RANS evaluations it saves on the way to that plateau, not the evaluations it skips. Where a final value is quoted instead, the text says so.

The starting point is the first version of the optimizer, arm A of Section 3.5, expected improvement on the twenty two CST coefficients with every proposal verified. It is a standard Gaussian process on the standard parametrization, the configuration one obtains before any modelling choice is made. Figure 5.1 shows it in grey on the nine cases, together with the two arms that follow it. It never reaches ninety eight per cent of the plateau of the reference loop within the budget. Everything that follows measures where that gap comes from, the search space in Section 5.2, the acquisition in Section 5.3, the prior mean in Section 5.4.

5.2 Searching a smaller space: the POD latent

One thing changes from arm A to arm A′. The twenty two CST coefficients become the eight POD coordinates of Section 2.3, with the same acquisition, the same box, the same manufacturability rules and the same verification of every proposal. From grey to blue in Figure 5.1 is the first large step of this chapter. It appears on all seven lift to drag cases and on both drag cases, and on all nine the blue curve reaches within the budget a plateau the grey curve does not approach.

The reason is the regime of the Gaussian process. With twenty two dimensions and a hundred evaluations the process has far fewer points than it needs to set one length scale per dimension, its posterior stays close to its prior over most of the box, and the acquisition proposes designs that the data do not support. With eight dimensions the same hundred points are dense enough for the length scales to be learned, the posterior mean has a shape, and the acquisition proposes designs on that shape. The reduction is a restriction, Section 2.3 measured what it leaves on the shape, and the

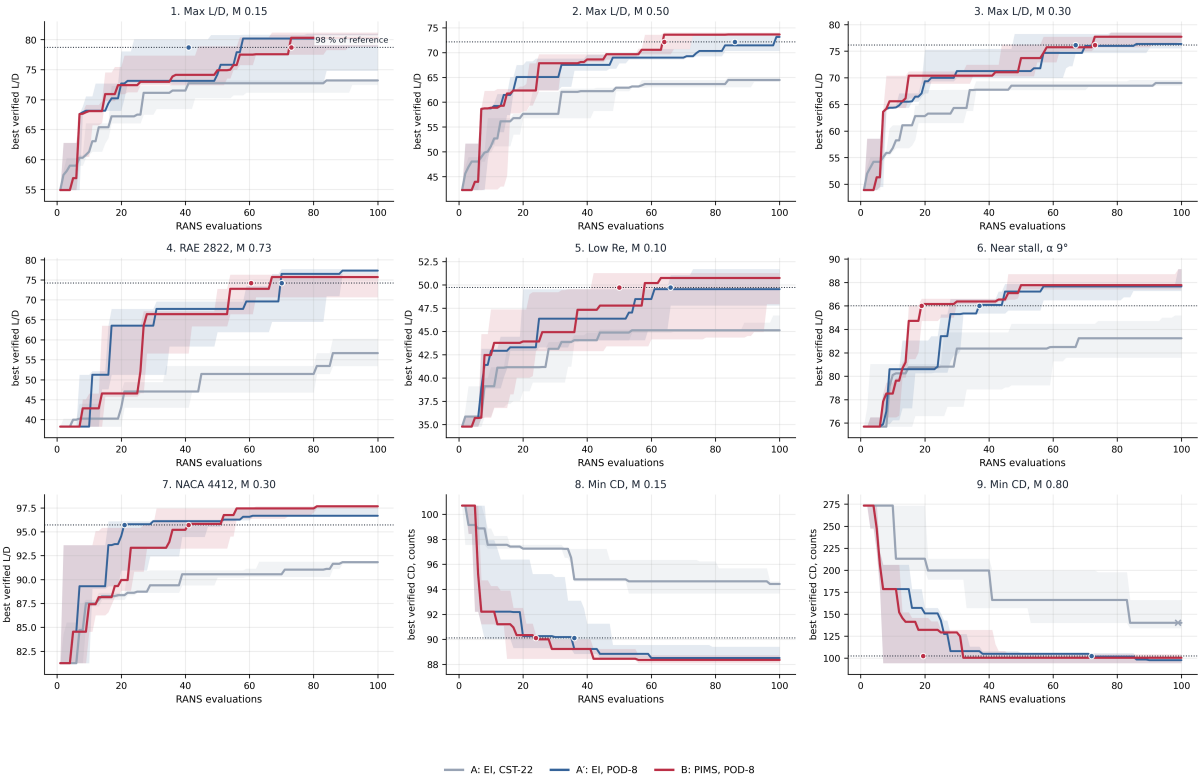


Figure 5.1: One lever at a time on the nine cases of Table 3.1, every design verified, median of three seeds with the band between seeds. Grey, arm A, expected improvement on the twenty two CST coefficients. Blue, arm A′, the same acquisition on the eight POD coordinates. Red, arm B, PIMS on the eight POD coordinates, the reference loop. From grey to blue only the search space changes, from blue to red only the acquisition. The dotted line is ninety eight per cent of the plateau of arm B and a dot marks where a curve crosses it. Blue often lies under red, the two arms reaching the same plateau. Panels follow the numbering of Table 3.1, the last two minimise drag.

question is whether the restriction costs performance. It does not, the best verified designs of arm A′ reach at least what arm A reaches on the nine cases where the two are separable. The latent does not reproduce the design arm A delivers, it finds a different and better one. The latent space is also an implicit shape constraint, a design expressed on eight modes is a plausible airfoil, and Chapter 4 showed that such constraints bound the optimism of the surrogate.

This step is not an economy of RANS evaluations in the sense of Chapter 4, every design is still verified. It is a better use of the same budget, and it is the step on which the two others are built, since the acquisition of Section 5.3 and the prior of Section 5.4 both act on a Gaussian process that has a shape to work with.

5.3 Placing the next point better: the acquisition

Two families. Once the search space is fixed, the acquisition decides where the next RANS evaluation goes. Two families exist. A global acquisition scores every candidate of the pool from the posterior of the Gaussian process, μ_n and s_n in Section 3.5, and lets the predicted variance drive the exploration. A local acquisition takes the maximum of the posterior mean inside a region around the current best verified design, evaluates it, and lets the agreement between prediction and measurement, not a predicted variance, decide how far the next step may go. This is the trust region model management (TRMM) of March and Willcox [28], and it is the second lever of this chapter. Five global acquisitions are implemented in the loop, and all of them share the pool of Section 3.5, two thousand

manufacturable candidates redrawn at every iteration, so that the acquisition is a discrete argmax and never a continuous optimization. All work in the standardised units of the process, with y_{best} the best verified value, $z(x) = (\mu_n(x) - y_{\text{best}})/s_n(x)$, and Φ, φ the normal distribution and density.

The global acquisitions as implemented. Five are implemented, all on the same pool and in the same standardised units. Table 5.1 gives each one and the constraint it carries in our setting. Two of them need a word, because a property of theirs is used elsewhere in this thesis.

Table 5.1: The five global acquisitions, with $z = (\mu_n - y_{\text{best}})/s_n$ and Φ, φ the normal distribution and density. Sampled quantities use one joint posterior path for PIMS, twelve for MES, and a subset of two hundred and fifty six candidates for the two sampled ones and for the knowledge gradient.

acquisition	score of a candidate	constraint here
EI, expected improvement [22]	$(\mu_n - y_{\text{best}})\Phi(z) + s_n\varphi(z)$	flattens far from the data, measured below
UCB, upper confidence bound [42]	$\mu_n + \kappa s_n$, $\kappa = 2$	the regret bound wants κ to grow with the iteration, we keep it fixed
PIMS, improvement over a sampled maximum [43]	$\Phi\big((\mu_n - \tilde{m})/s_n\big)$, $\tilde{m}$ the max of one path	the threshold moves every iteration, ties saturate
MES, max value entropy search [45]	mean over twelve paths of $\frac{g\varphi(g)}{2\Phi(g)} - \log\Phi(g)$	same sampling noise as PIMS, averaged
KG, knowledge gradient [15]	expected gain of the posterior maximum after the evaluation	cost quadratic in the subset, so the subset is small and the estimate noisy

Expected improvement is the standard of the field and the acquisition of the first baseline, and its constraint in our setting is numerical rather than conceptual. Far from the data both of its terms vanish, the flattening documented by Ament et al. [4], and we measure it in our own runs. Figure A.6 in the appendix records, at every iteration of arms A and A′, the largest expected improvement found in the pool. In the twenty two dimensional box it sits a hundred to a thousand times below the latent value over most of the budget, so the argmax is taken among candidates that all promise next to nothing. In the eight mode latent the pool stays dense enough for the acquisition to keep a signal. This is the numerical half of the gap that Section 5.2 attributes to the search space.

PIMS is the acquisition of the reference loop, and the reason is a property that every paired comparison in this thesis relies on. Its sample path is seeded by the iteration index, so two arms that have seen the same data see exactly the same acquisition, and a difference between them cannot come from the draw. The low Reynolds panel of Figure 4.2, where two arms coincide exactly because the gate never fires, is that property visible.

Why the zoo cannot enter the acquisition. The multi source extension of the knowledge gradient [32] puts every information source in one joint process and buys the evaluation with the best information per unit cost. It is the natural way to let a zoo enter the acquisition rather than the prior, and it is where the assumption of Section 2.4 bites. That extension needs sources whose bias against the truth is consistent and smooth, a coarse mesh and a fine one, and our zoo offers no such ladder. We tested it on a toy problem and on case 1: a cheap source with an inconsistent bias degrades the joint process instead of informing it. It is not pursued further, and it is the reason the zoo appears in this thesis as a prior mean and never as a fidelity.

Global acquisitions compared. Two of them are compared on the nine cases, expected improvement and PIMS, arms A′ and B of Figure 5.1, from blue to red. Where they arrive is close: on the seven lift to drag cases the final values differ by less than two units, and on the transonic drag case expected improvement ends three counts lower. How fast they get there is not: PIMS reaches the target sooner

on six cases, expected improvement on cases 1 and 7, and the two tie on case 4. This agrees with what the literature reports once expected improvement is computed without its numerical flattening [4].

The five are compared on case 1 in Figure 5.3. They end within two units of lift to drag ratio of one another, so the acquisition does not decide where the search arrives. They differ in how fast they get there, and the ordering is not the one the literature would suggest: the cheapest to the target are expected improvement and the knowledge gradient, the dearest are the upper confidence bound and PIMS, and the spread between the fastest and the slowest is close to a factor of two. Two remarks bound what this measures. It is one case and three seeds, so the ordering is indicative and we do not generalise it. And the target is the plateau of the reference loop, which is PIMS, so PIMS is not favoured by the choice of target. Expected improvement and the knowledge gradient are the fastest on this case, but it is one case, the seeds of the knowledge gradient reach the target anywhere between nineteen and eighty evaluations, and with an estimate that is itself noisy since it is computed on a small subset of the pool. And once a prior is used, the global acquisition is not what the method runs: the trust region step replaces it, and Section 5.5 shows that this matters more than which global acquisition is chosen. The campaign of this chapter therefore does not select an acquisition on speed. It keeps PIMS for the reference loop because it has no parameter and because its fixed path seed gives two arms the same acquisition at equal data, the property every paired comparison here relies on, expected improvement for the first baseline because it is the standard of the field, and the trust region step for the arms with a prior, for the reason given below.

The trust region step. The local acquisition replaces steps 4 to 6 of Algorithm 1. Let x_{best} be the current best verified design and Δ the radius of the region, in the unit latent cube, with $\Delta_0 = 0.4$ at the start. Let m_k be the model of the objective at iteration k, the posterior mean of the Gaussian process here, and the debiased surrogate corrected by the Gaussian process in Section 5.4. The step is

$$x_k = \arg\max_{\|x - x_{\text{best}}\|_\infty \le \Delta} m_k(x), \qquad \rho_k = \frac{y(x_k) - y(x_{\text{best}})}{m_k(x_k) - m_k(x_{\text{best}})}, \qquad \Delta \leftarrow \begin{cases} \min(2\Delta, 1) & \rho_k \ge 0.2, \\ \Delta/2 & \text{otherwise,} \end{cases}$$

where y is the RANS value. The acceptance threshold and the growth and contraction factors are the values published by March and Willcox [28]. The initial radius, four tenths of the unit cube, and the collapse threshold are ours, their paper scales both to the problem, and neither was tuned in this thesis. The candidate is evaluated with RANS at every iteration, so every iteration costs one evaluation and adds one anchor, and x_{best} moves only when the measured value improves. The ratio ρ_k compares the improvement measured to the improvement the model predicted. When the model kept its promise the region grows, when it over promised the region shrinks, which is also a protection against the optimism of Chapter 4, the step is never larger than what the last measurement justified. No variance enters the decision, so the exploration comes from what RANS confirmed and not from what the process believes about itself. Figure 5.2 draws the step and the life of the radius along a run.

Restart at the plateau. The region shrinks at every step whose measured gain falls short of the prediction, and when its radius falls below one per cent of the cube the model has nothing left to propose around x_{best} that RANS confirms. The original method stops there, and in our runs that happens early: without a restart and without a prior mean the region collapses after sixteen to twenty three RANS evaluations on cases 1 to 3, at a value the global acquisitions pass well before the end of their budget. Without a restart the trust region step is a phase, not a method. The restart rule used in the rest of this chapter resets the radius to Δ_0 whenever it collapses, as long as budget remains. The centre stays the best verified design, the process keeps every anchor, and the next candidate is again the maximum of the model, now inside the wider region. When the arm carries a prior mean, Section 5.4, the restart touches only the radius: the prior is the one the arm uses at that iteration, fixed in Section 5.4 and chosen online in Section 5.6, and the model that fills the reopened region is that prior corrected by the anchors gathered so far. The restart therefore removes the first constraint of the step, the premature stop, and every trust region arm of Sections 5.4 to 5.7 carries it. It does not

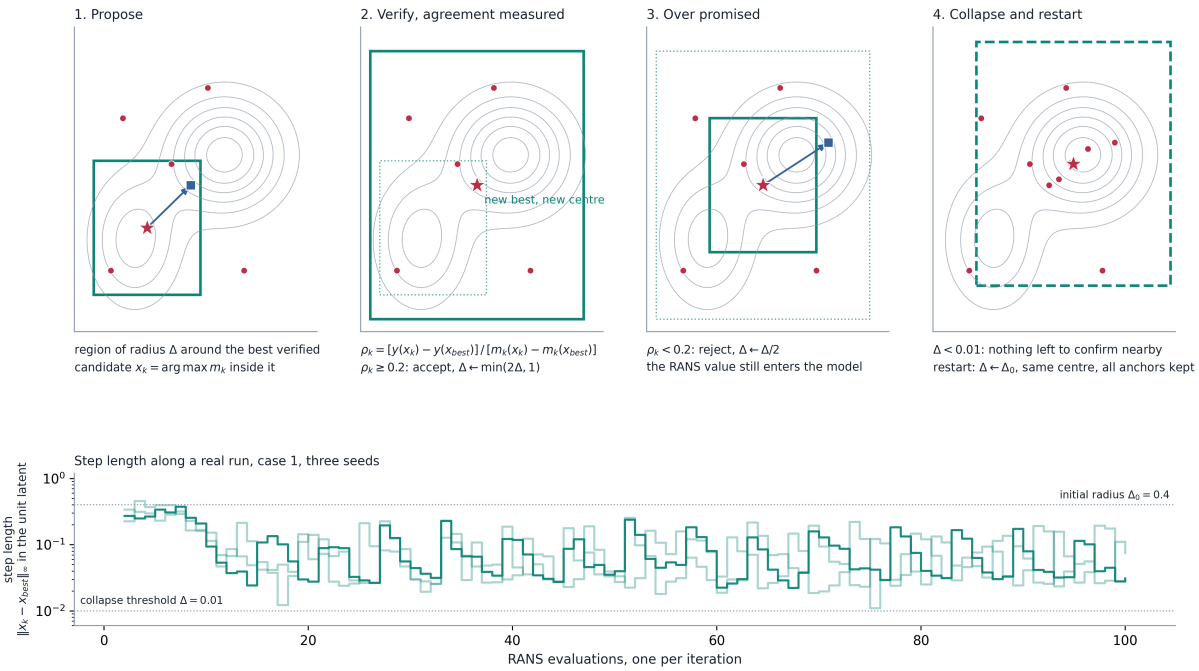


Figure 5.2: The trust region step. Top, four moments of the step on an illustrative landscape, the red star is the best verified design, red points are RANS anchors, the green box is the region, the blue square the candidate at the maximum of the model inside it. Bottom, the radius along an illustrative run, doubled at every accepted step, halved at every rejected one, reset to its initial value when it falls below the collapse threshold.

remove the second. Nothing in the restart sends the search elsewhere, only the shape of the model inside the reopened region can. Figure 5.3 makes the point on case 1: with the restart the step now spends its whole budget, and it still never reaches the plateau of the reference loop, ending two and a half units below it. The restart buys persistence, not direction. The shape that supplies the direction is the subject of the next section.

Why local, and why not alone. At small budgets the mean of the model is a better guide than its variance. A greedy step along the mean, bounded by a region that shrinks when the model is wrong, spends each evaluation where the model expects a gain and learns from the answer, whereas a variance driven acquisition spends part of the budget reducing uncertainty in places that never matter. This is the ordering the literature reports at small budget and in high dimension [11]. It is not what we measure without a prior mean. In Figure 5.3 the trust region step is the slowest arm over the whole budget and the only one that never reaches the target. The reason is the second constraint above. A posterior mean fitted on six anchors has a shape only near those anchors, so the maximum inside the region is found in a few steps, the region closes, and the restart reopens it around the same information. The local step is not a weaker acquisition, it is an acquisition with nothing to be greedy about. What it lacks is a model with a shape beyond the anchors, and that is what the prior mean of the next section supplies.

5.4 The surrogate as prior mean of the Gaussian process

The mechanism. The third lever gives the surrogate a third role. It is neither a substitute for an evaluation, Chapter 4, nor a candidate for the acquisition, Section 5.3. It is the prior mean of the Gaussian process. With f_{low} a member of the zoo chosen for the case, $\hat{\delta}$ its residual model of Section 2.5, and e

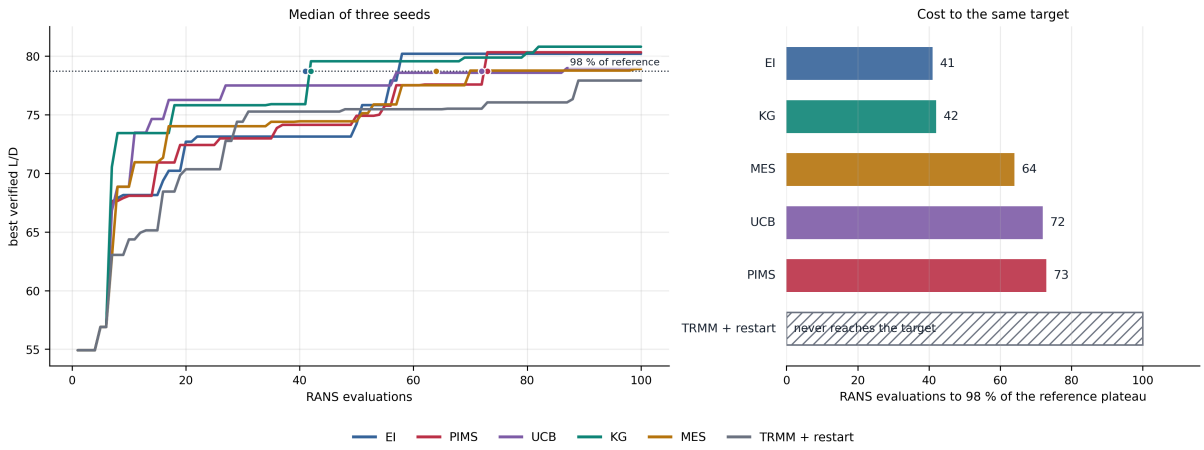


Figure 5.3: Acquisitions at equal search space, case 1, the eight mode latent, every proposal verified, median of three seeds. No arm carries a prior mean. *Left:* the five global acquisitions and the trust region step with restart; the dotted line is ninety eight per cent of the plateau of the reference loop and a dot marks the crossing. *Right:* the RANS evaluations each needs to reach that target, an open hatched bar meaning the target is never reached. The five global acquisitions end within two units of one another and differ by close to a factor of two in cost; the trust region step without a prior never gets there.

the Gaussian process, the model of the objective is

$$m(x) = f_{\text{low}}(x) + \hat{\delta}(x) + e(x), \qquad e \sim \mathcal{GP}\big(0, k\big) \text{ conditioned on } y_i - f_{\text{low}}(x_i) - \hat{\delta}(x_i) \text{ at the anchors.}$$

The process no longer carries the objective, it carries the gap between the corrected surrogate and RANS. If that gap is small and smooth, the six anchors of the initialisation are enough to fix it, the posterior mean has the shape of the surrogate everywhere and the value of RANS near the anchors, and the maximum of the model inside the trust region falls near the true optimum from the first iterations. Nothing else changes. The step is the one of Section 5.3, with m_k the model above, one RANS evaluation per iteration, the ratio ρ_k deciding the radius, the restart at the collapse.

What it changes for the local step. Section 5.3 left the trust region step with a model that has a shape only near its anchors, so that a reopened region proposes again what the anchors already know. With the prior the model has a shape everywhere in the box. The step goes where the surrogate says the optimum is, RANS confirms or refuses, and ρ_k tightens the region as soon as the surrogate over promised. The surrogate therefore returns as a guide and never as a value. No point enters the process without a RANS evaluation, which is the exact difference with the router of Chapter 4, and the optimism of the surrogate at its own optimum, Section 4.3, costs at most the evaluations it takes to refuse it.

Result on the nine cases. The prior is held fixed in this section, the NeuralFoil member corrected by its residual model on every case, so that the lever is measured without any selection rule. Section 5.6 takes up the choice. Figure 5.4 compares four arms on the nine cases, the reference loop, the same trust region step and restart without a prior mean, and with it, the prior being either NeuralFoil named by hand or the member chosen online by the selector of Section 5.6. The arm without a prior is the control that separates the local step from the prior, and it is why it is drawn.

On the six subsonic lift to drag cases the prior arm reaches ninety eight per cent of the plateau of the reference loop in ten to twelve RANS evaluations, where the reference loop needs nineteen to seventy three. The control without a prior reaches it on the median only on the near stall case and

on NACA 4412, in twenty two and thirty evaluations; on the four others at least two seeds out of three never reach it. On those cases the economy is the prior and not the local step.

The three transonic and drag cases behave differently and each in its own way. On the transonic off design case the restart alone is the cheapest arm, twenty five evaluations against sixty seven for the reference, and adding the prior makes it slower, forty two: the prior earns nothing there. On the transonic drag case the prior arm is the fastest, twenty one evaluations, but the restart alone already gets to twenty five, so the prior adds little. On the incompressible drag case the reference needs twenty four evaluations and the prior arm forty three. That case separates the two mechanisms this section conflates elsewhere. The prior is not what loses there: within the trust region arm it brings the cost from sixty nine evaluations without it to forty three with it. What loses is the local step itself, which the global acquisition beats on this case. The prior does its job and cannot rescue the machine it rides on.

Where the prior helps and where it does not. The separation is by objective and by regime. On the six lift to drag cases the prior reaches the target of the reference loop between one and a half and seven times sooner. On the three remaining cases, the two drag cases and the transonic off design case, it gives no clear gain over the best arm without it.

What we cannot do is explain the separation by a property of the prior field itself. The natural candidate is the size of the residual the process is then left to fit: measure, at the designs the optimizer verified, the spread of truth minus prior, and divide it by the spread of the truth among those same designs. Below one the prior leaves an easier problem than the truth itself, above one a harder one, and the quantity costs no extra RANS evaluation. Measured on designs taken only from arms that used no prior at all, so that the population is not one a prior already steered, it does not predict the outcome: the low Reynolds case sits at one and two tenths and is almost six times faster, the cambered case at one and three tenths and three and a half times faster, both on the wrong side of the threshold, and the rank correlation between the quantity and the speed-up is -0.40 over the nine cases. We report it because it is the first thing a reader will propose, and because it is measurable and wrong. The regime predicts, the residual does not, and we have no mechanism that explains why beyond the reliability map of Chapter 2.

The residual correction $\hat{\delta}$ sits under the same uncertainty, and we measured it by running the same arm with the raw member, seed by seed, on the nine cases. On twenty seven paired runs the correction reaches the target sooner on sixteen, ties on five and is slower on six. On five runs the raw member never reaches the target and the corrected one does; the reverse happens once, on one seed of case 8. Case by case, the correction is faster on seven of the nine, by a wide margin on cases 1, 2, 3 and 9, where the raw member mostly fails, and by a few evaluations on cases 5, 7 and 8. It is slower on case 4, forty two evaluations against thirty two, and on case 6, eleven against seven. When it does not help it costs little: the slower runs lose two to thirteen evaluations, apart from the one seed of case 8. We therefore keep $\hat{\delta}$ as a component whose value is measured, not as a contribution: it helps where the raw member is biased in a way the probe captured, and it cannot supply what the probe did not see, Section 2.5.

What the prior is not, and why there is no gate here. It is not an economy by trust. Every design this arm proposes is verified with RANS, the tolerance of Chapter 4 is set to zero and step 6 of Algorithm 1 never fires. That is deliberate and it is the difference that matters. The gate tried to buy economy by skipping evaluations, and Chapter 4 showed it cannot. The prior buys economy by changing where the evaluations go: the model has a shape everywhere in the box from the first iteration, so the search spends its budget near the optimum instead of discovering the shape. The surrogate is worth more as a shape that is always verified than as a value that is sometimes believed, and that sentence is the result of this thesis in one line. It is not a fusion of the zoo into the acquisition, which Section 5.3 set aside. And it is not yet a choice, the member is fixed here and the rule that picks it is the object of Section 5.6. What remains to settle first is the acquisition that goes with a prior, the global one or the

local step, which is the next section.

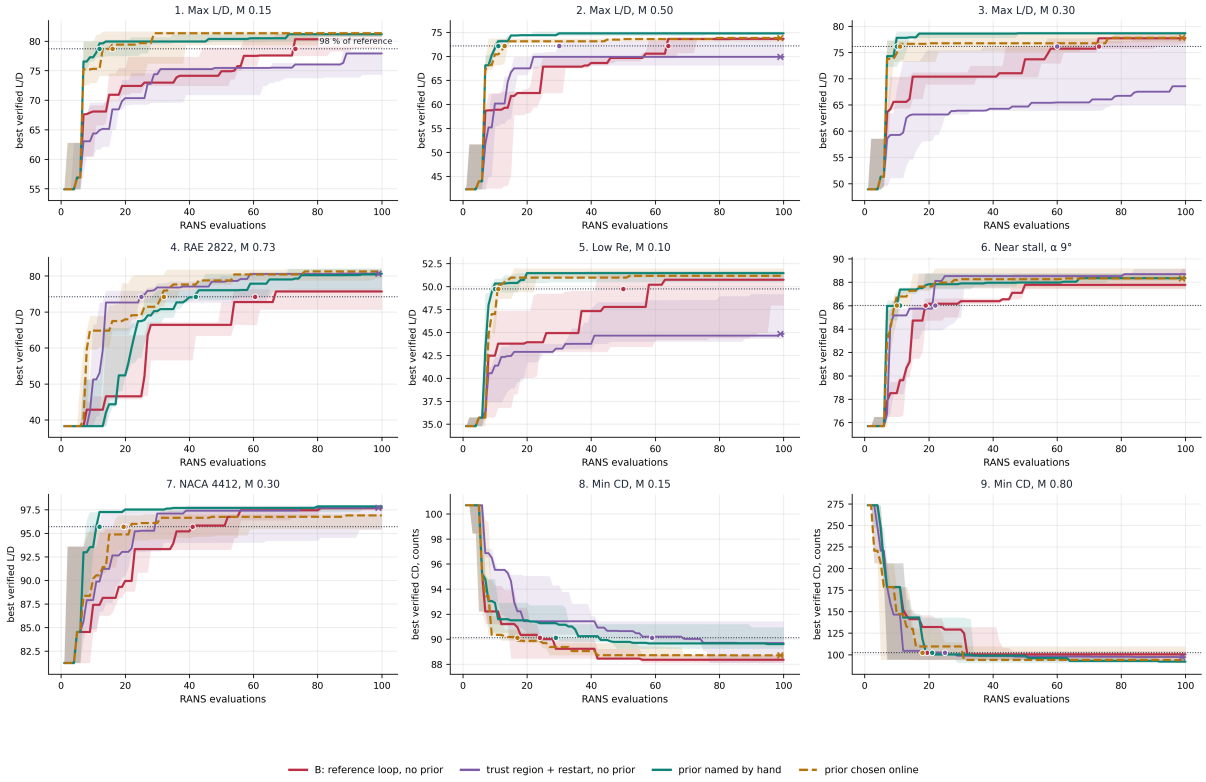


Figure 5.4: The surrogate as prior mean on the nine cases of Table 3.1, median of three seeds with the band between seeds. Red, the reference loop B. Purple, the trust region step with restart and no prior mean, the control that separates the local step from the prior. Green, the same arm with the NeuralFoil member, corrected by its residual model, as prior mean. Dashed, the same arm with the prior chosen online among the zoo, Section 5.6. The dotted line is ninety eight per cent of the plateau of the reference loop and a dot marks where each arm crosses it. A cross marks where the first seed of an arm ended.

What the optimizer actually drew. Figure 5.5 shows the designs the loop delivers, and they are the ones an aerodynamicist would draw. On the lift to drag cases both arms add camber and pull the lower surface up; on the transonic case the section gets thinner, which weakens the shock; on the two drag cases the section stays thin and close to symmetric. No shape is wavy, the manufacturability rules hold. This validates the harness; it is not a result.

The prior arm does not reach the same optimum sooner, it reaches a different one. On eight of the nine cases it moves further from the starting airfoil than the baseline, up to seven point eight per cent of chord against five point two on case 7. The exception is instructive: on case 8 the baseline moves more, six point three per cent against four point four, for a worse design, ninety four counts against ninety. What counts is the direction, not the amplitude.

The prior arm also delivers the same airfoil from one seed to the next, Figure 5.6. The largest gap between the three seeds is one point two per cent of chord in median against three point seven for the baseline, and the ordering holds on all nine cases. For the baseline that gap is of the order of the displacement itself, each seed delivers its own airfoil. This argument is independent of speed: at equal value one prefers the method that always delivers the same design. Case 4 is the least reproducible, two point four per cent. Part of the convergence comes from the search space, which the box and the manufacturability rules already make narrow.

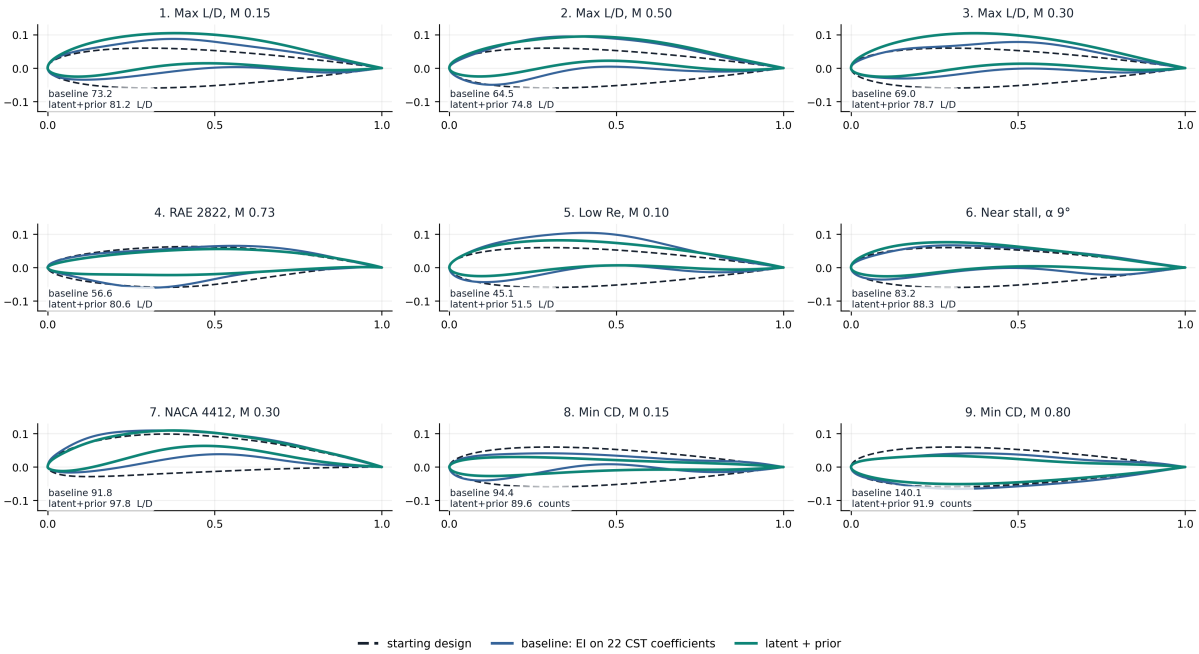


Figure 5.5: The delivered designs on the nine cases of Table 3.1, decoded from their CST coefficients. Dashed, the starting airfoil. Blue, the baseline, expected improvement on the twenty two coefficients. Green, the same loop in the eight mode latent with the surrogate as prior mean, NeuralFoil named by hand. Each curve is the best verified design of the median seed.

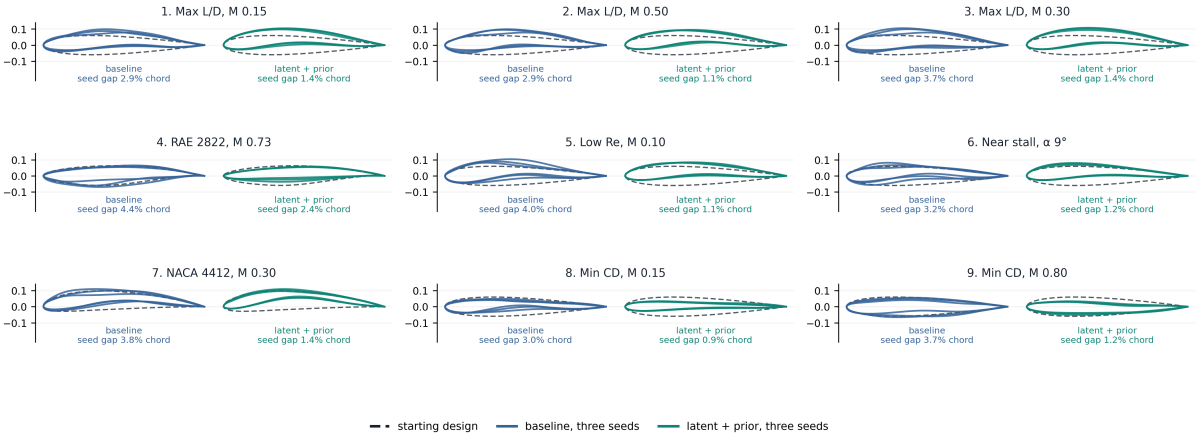


Figure 5.6: Reproducibility of the delivered design, the two arms side by side, three seeds each, over the dashed starting airfoil. The baseline draws three visibly different airfoils, the latent and prior arm draws three that nearly coincide: a median gap between seeds of three point seven per cent of chord against one point two. The ordering holds on all nine cases.

5.5 Acquisition × prior: the 2×2

Figure 5.7 separates the two mechanisms of the prior arm on the three cases where every cell ran. Without a prior mean, the trust region step with restart is the slowest arm there is: it reaches the target on at most one seed out of three, where the reference acquisition over the whole box reaches it on all three in sixty four to seventy three evaluations. The prior alone, with the reference acquisition, already brings that to eighteen to twenty eight. The two together bring it to ten to twelve. The prior is the dominant factor, and the local step pays only once the prior gives the model a shape to follow; without one, the step has nothing to be greedy about. The same holds with the prior chosen online,

at a higher cost when the trust region is absent, twenty four to forty six evaluations against eleven to sixteen with it.

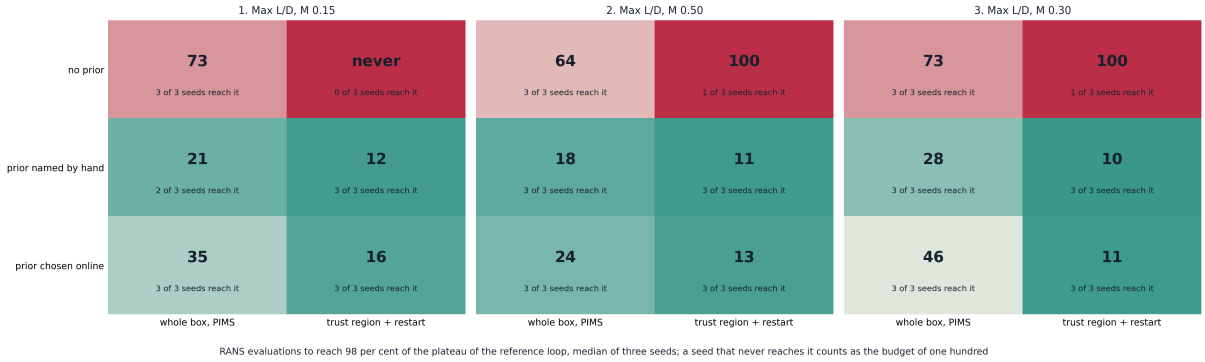


Figure 5.7: The acquisition step against the prior, three cases. Rows: no prior mean, NeuralFoil named by hand as prior mean, the prior chosen online. Columns: the reference acquisition over the whole box, or the trust region step with restart. Each cell is the median over three seeds of the RANS evaluations to reach ninety eight per cent of the plateau of the reference loop, a seed that never reaches it counting as the budget, with the number of seeds that do reach it.

5.6 Choosing the prior

Every result of Section 5.4 was obtained with a prior we named ourselves. That is the weakest part of the method, and it is the part this section removes. The question is whether the optimizer can choose its own prior from the zoo, using only what it has already paid for.

The choice is worth more than the mechanism. Figure 5.8 runs the same loop on the same case and changes one thing, which member is used as the prior mean. On the lift to drag case at Mach 0.15 the best member ends at 81.2 and the worst at 63.6, a spread of seventeen and a half in lift to drag, where the prior itself is worth nine tenths of a point of plateau over a loop that uses none. The comparison that matters is not with the loop that has no prior but with the loop that has the wrong one: a badly chosen member ends seventeen below it, and never reaches the threshold the plain loop reaches in seventy three evaluations. The same ordering holds at Mach 0.30, nine and a half of spread, and at Mach 0.50, three and a half.

The fourth panel is the same statement seen from the other side. On the minimum drag case the member that wins the first three, NeuralFoil, is the wrong one: it ends at 89.6 counts against 88.3 for the loop that uses no prior, and it is above that loop for the whole run. So the sign of the lever is not a property of the member either. A member is not good or bad, it is good or bad on a case, and no fixed choice covers the set. This is why selection is not a refinement of Section 5.4 but the condition under which its result can be used at all, since nothing in that section tells a practitioner which member to name on a case they have not run.

Ranking, not accuracy. Every offline criterion we tried failed at this: the calibrated uncertainty of Section 2.5, the accuracy on the probe set, the agreement with a single anchor. They all score a member by how close it is to the truth, and Chapter 4 is the reason they fail, since closeness measured anywhere other than where the optimizer goes does not survive the trip. What works is to score a member by the *order* it puts on the designs already verified. Following the ranking weighted ensembles of Feurer et al. [14], each candidate m is charged with the number of pairs of anchors whose order it contradicts,

$$\ell_m \;=\; \left|\{(j,k),\, j<k \;:\; \operatorname{sign}\big(b_m(x_j)-b_m(x_k)\big) \neq \operatorname{sign}\big(y_j-y_k\big)\}\right|,$$

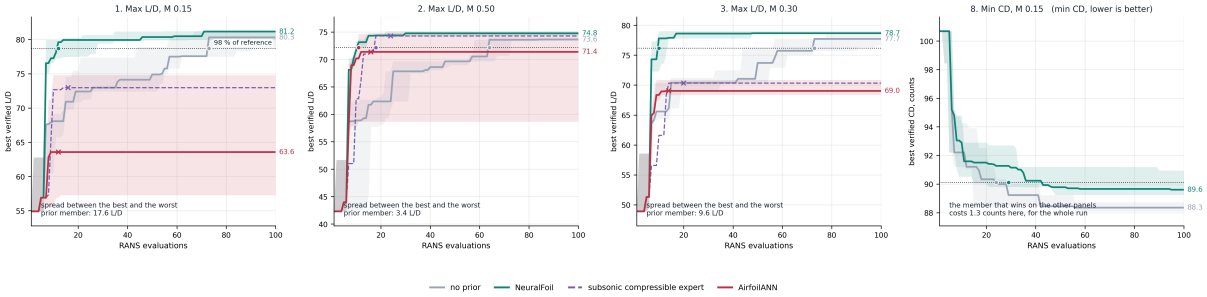


Figure 5.8: One loop, one case, one difference: the member used as the prior mean of the Gaussian process. Median of three seeds with the band between seeds, except the subsonic compressible member on the first three panels, a single run drawn thinner. The fourth panel is an objective (ii) case, drawn in drag counts, where lower is better. The dotted line is ninety eight per cent of the plateau of the reference loop and a dot marks where a curve crosses it. A cross marks a run that ended before the budget. The spread between the best and the worst member is wider than what the prior buys over using none.

with b_m the corrected belief of member m and y the verified value. Ties count as no contradiction. The anchors are then resampled with replacement a hundred times, and the weight w_m is the fraction of draws in which m carries the smallest charge, a tie splitting the draw. The corrected prior mean $f_{\text{low}} + \hat{\delta}$ of Section 5.4 is replaced by the mixture

$$f_{\text{prior}}(x) = \sum_m w_m \, b_m(x), \qquad \sum_m w_m \le 1,$$

and the Gaussian process is fitted on $y_i - f_{\text{prior}}(x_i)$. The weights are recomputed at every iteration, on all the anchors paid for so far, at no RANS cost. The two constants, the number of draws and the dilution threshold below, are those of Feurer et al.; neither was tuned here.

The right to abstain. One candidate is not a member. It is the Gaussian process fitted on the RANS values alone, with no prior mean, scored by its leave one out prediction at each anchor, which has a closed form [36],

$$\mu_{-i} = y_i - \frac{\left[K^{-1} y\right]_i}{\left[K^{-1}\right]_{ii}},$$

with K the covariance matrix of the anchors, noise included. It stands for the decision to use no prior at all, and it is charged like any member. Its weight enters the mixture multiplied by zero, so weight moved onto it shrinks the prior rather than switching it off, which is why the weights sum to at most one. It also serves as the bar every member must clear: a member whose median charge over the draws exceeds the ninety fifth percentile of the charges of this candidate is set to zero, the dilution rule of Feurer et al., so a member that orders the anchors worse than the process alone never enters the prior. Algorithm 2 gathers the steps and Figure 5.9 draws one turn of the loop.

As anchors accumulate the abstention weight rises to one on every run whose weights we could read, by twenty six anchors, and by thirty six on case 9. The weight on the members is largest at six anchors, already reduced at sixteen, and gone at twenty six: the prior acts where it is useful, early, and retires by itself once the process has enough data of its own.

Algorithm 2. Online choice of the prior, run at every iteration of the trust region step, before the Gaussian process is fitted. It costs no RANS evaluation.

1. Take the n anchors (x_i, y_i), every design verified with RANS so far.
2. For every member m of the zoo, compute its corrected belief $b_m(x_i)$ at the anchors.
3. Add the candidate *no prior*: the leave one out mean μ_{-i} of the Gaussian process fitted on y alone.

4. Repeat a hundred times: draw n anchors with replacement, compute the charge ℓ of every candidate on the draw, credit the candidate with the smallest charge, ties sharing the credit.
5. Set w to the share of draws each candidate won.
6. Set to zero the weight of every member whose median charge exceeds the ninety fifth percentile of the charges of *no prior*, and renormalise.
7. Build the prior mean $f_{\text{prior}} = \sum_m w_m b_m$, the weight on *no prior* adding nothing.
8. Fit the Gaussian process on $y_i - f_{\text{prior}}(x_i)$, take the trust region step on $f_{\text{prior}} + e$, verify the proposal with RANS; it becomes the anchor $n+1$.

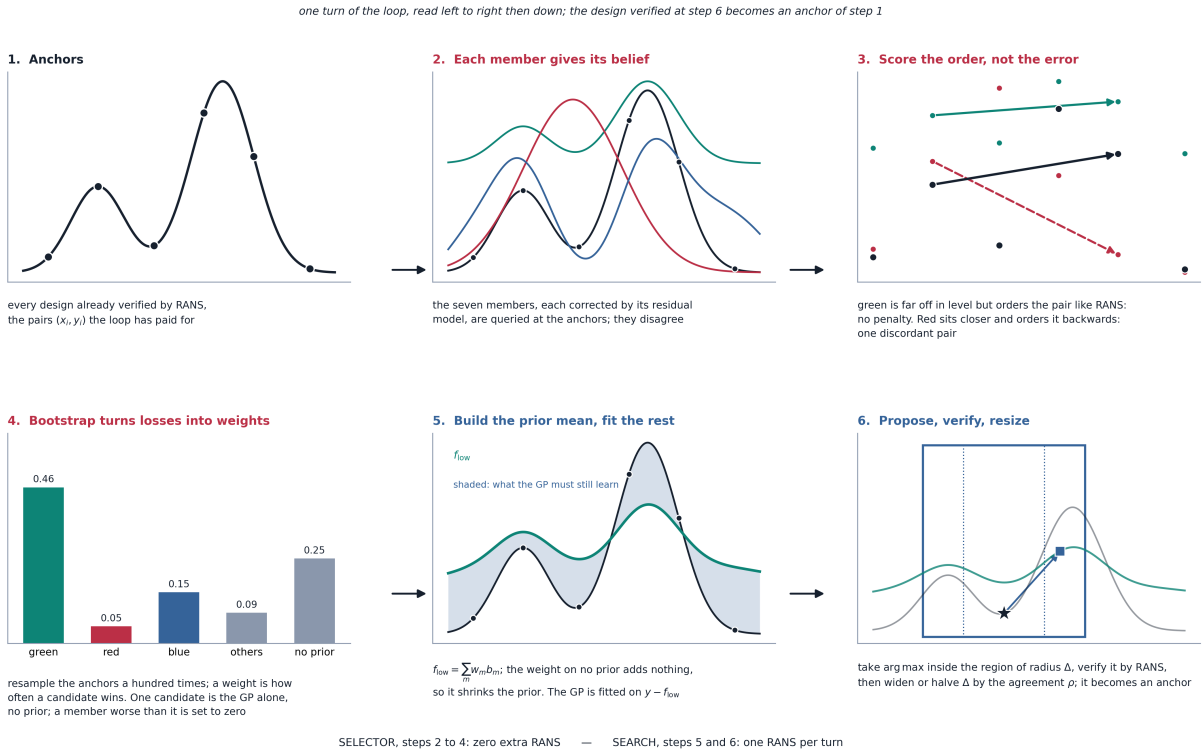


Figure 5.9: One turn of the loop with the prior chosen online, on an illustrative one dimensional objective. Steps 2 to 4 are the selector and cost no RANS evaluation; steps 5 and 6 are the trust region step of Section 5.3, one RANS evaluation per turn. The member that wins is not the one closest to the truth but the one that orders the anchors like it.

What it elects. At six anchors, on the subsonic lift to drag cases, the heaviest member is most often NeuralFoil, with a third to a half of the weight, the member Section 5.4 had chosen by hand. The rest of the weight is spread, and by sixteen anchors abstention holds most of it on almost every run. The prior therefore acts early and briefly, which is where Section 5.4 located its value.

What choosing is worth. Measured against the loop with no prior, the selector reaches the target sooner on all nine cases, sixteen evaluations on the median against fifty eight. Measured against NeuralFoil named by hand, it is within one to four evaluations on five cases, cases 1, 2, 3, 5 and 6. It is faster on the two cases where NeuralFoil is the wrong choice, by ten evaluations on the transonic off design case and by twenty six on the incompressible minimum drag case, where the member named by hand is slower than using no prior at all. It removes the human choice and does not pay for it, except where the next paragraph says. Read against Figure 5.8, that is the right accounting. What selection buys is not the nine tenths of a point the prior adds over no prior, it is the seventeen it costs to name the wrong member, and that is a loss no amount of tuning elsewhere recovers.

Where it falls short. On cases 7 and 9 the selector is ten evaluations slower than NeuralFoil named by hand. At six anchors, on most seeds, no member orders the anchors clearly better than the others, the weight is spread, and it moves to abstention within sixteen anchors: the prior that would have helped is dropped before it has proven itself. Six anchors are a small sample to rank on, and this is the price of refusing to commit on it.

Two variants we rejected. Holding the prior off for the first anchors and switching the ranker on later, so that its decision rests on more data, is worse on cases 1 to 3, because it deletes the window in which the prior does its work. Weighting every member equally, which removes the ranking entirely, is the fastest arm of all on one of these cases and collapses on another, below the loop with no prior at all. The ranking is what keeps the prior from being a gamble.

5.7 Putting it together: cumulative gain over the baseline

Figure 5.10 puts the levers of this chapter end to end, one mechanism per rung, on the nine cases. The number is the median over the cases of the RANS evaluations needed to reach ninety eight per cent of the plateau of the reference loop, a seed that never reaches it counting as the budget. The first baseline, expected improvement on the twenty two coefficients, never reaches it. The latent space brings the median to seventy, the acquisition to fifty eight, and the prior chosen online to sixteen, the largest single step. Naming the right member by hand would bring it to twelve: removing the human choice costs four evaluations on the median. The right panel keeps the median honest. The first baseline fails on every case, and the final arm reaches the target on every case, on all three seeds except on cases 7 and 9, where one seed misses it. The curves behind these numbers are those of Figure 5.4.

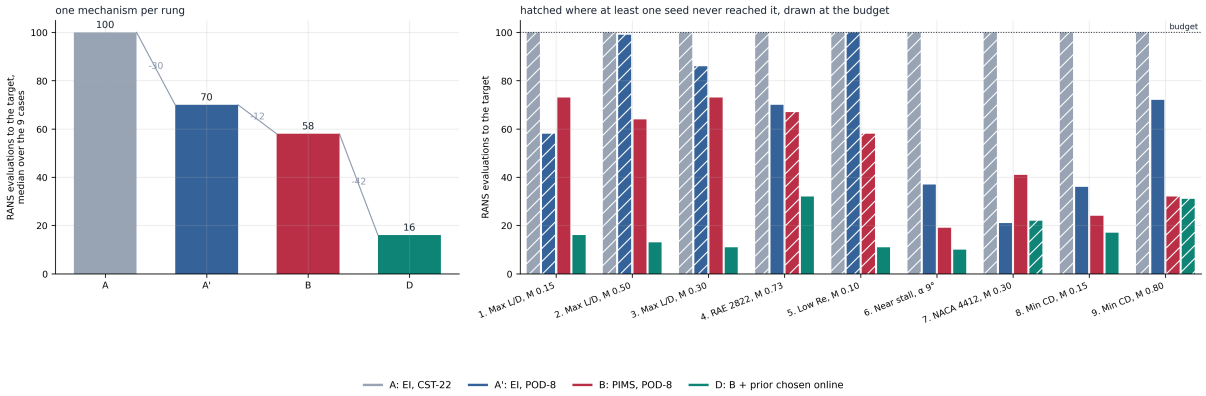


Figure 5.10: Where the economy is bought, one mechanism per rung. *Left:* median over the nine cases of the RANS evaluations to reach ninety eight per cent of the plateau of the reference loop, a seed that never reaches it counting as the budget. A, expected improvement on the twenty two CST coefficients; A′, the same on the eight POD coordinates; B, PIMS on the eight coordinates, the reference loop; D, B with the prior chosen online inside a trust region with restart. *Right:* the same number case by case, hatched where at least one seed never reaches the target.

Chapter 6

Conclusion

6.1 What this thesis shows

The gate does not save. A gate that trusts a zoo member when its uncertainty is small, and pays for RANS otherwise, does not reach the plateau of verifying everything any sooner, on any of the nine cases. Where it trusts little it changes nothing, where it trusts a lot it ends below. A gate that trusts at the same rate on proposals drawn at random does no worse. Hypothesis H1 is refuted.

The uncertainty is not what fails. At the designs the optimizer elects, the uncertainty ranks the error, with a rank correlation near one half, and in the constrained space its nominal ninety per cent band covers ninety two per cent of the errors. Hypothesis H2 holds. It does not help the gate, because of a selection effect: the designs worth trusting are the promising ones, where error and uncertainty are large together, and the designs the gate can trust lie where RANS has already been paid for, where a belief adds nothing. Choosing the member point by point by its uncertainty adds nothing over the best single member either, and hypothesis H3 is refuted.

The economy comes from elsewhere. With every design verified, the number of RANS evaluations needed to reach the plateau of the reference loop falls, in median over the nine cases, from beyond the budget for expected improvement on the twenty two coefficients, to seventy in the eight mode latent space, to fifty eight with the reference acquisition, and to sixteen with a zoo member as prior mean, chosen online, inside a trust region with restart. The prior is the largest single step. Naming the right member by hand would give twelve, so removing the human choice costs four evaluations, and it protects against the member that, named wrongly, ends seventeen units below.

One zoo, two questions. The same zoo fails as a gate and succeeds as a prior because the two roles ask it different questions. The gate asks for a magnitude, how far the member is from RANS at this design, and at the designs that matter no member knows. The prior asks for an order, which of two designs is better, and a member can order designs correctly while being far off in level. The surrogate is worth more as a shape that is always verified than as a value that is sometimes believed.

6.2 What uncertainty is actually for

The calibrated uncertainty of Section 2.5 is neither the gate nor the selector of the final method, which ranks members by the order they put on the anchors and never reads $\hat{\sigma}$. It keeps three uses. It draws the reliability map of the zoo, member by member and regime by regime, which tells a practitioner which members to admit before any optimization is run. It is the instrument that makes the negative result rigorous: because the uncertainty does rank the error at the elected designs, the failure of the gate cannot be blamed on a poor uncertainty model. And its companion, the residual correction $\hat{\delta}$, is

a component of the prior whose value is measured rather than assumed: it reaches the target sooner on seven of the nine cases and costs little where it does not.

6.3 Limits and threats to validity

The calibration reference is not the optimization reference. The residual and uncertainty models of Section 2.5 are fitted on a probe whose labels come from four solver families, and the loop verifies against our own SU2 setup. The smaller set recomputed with our solver, Section 2.2, judges the members; it does not fit the correction. This was a deliberate choice, and the alternative was worse: the largest source of the dataset supplies more than three quarters of its rows with one solver, and adopting that solver as the reference would have judged our own networks against the very code that produced their training labels, which would make the central result of Chapter 4 a test of self consistency. We bought independence of the reference and paid for it in the correction term. The price is measured. The member to reference bias differs between the two references by about four counts of drag for the global network and sixteen for NeuralFoil, against biases of order two and fifty counts, so the direction of the correction never changes and only its magnitude does.

The correction carries information only where the data differ. Three of the seven members, the global network and the two regime experts, are trained on the training split of the harmonised dataset the probe comes from, across all its sources, and their residual against its validation split is within two counts of drag. For them the correction has almost nothing to learn. It carries information for the members whose training data differ from the probe, the two source experts and the two external members, NeuralFoil and AirfoilANN, and Section 2.5 shows it changing their ordering of designs markedly. That the selector most often favours NeuralFoil, an external member, is consistent with this.

The reference is one acquisition among two. Every economy in Chapter 5 is counted against the reference loop, which uses PIMS. Expected improvement in the same space reaches the target sooner on some cases and later on others, Section 5.3, so neither dominates, and on case 1, the only case where all five acquisitions were run, expected improvement and the knowledge gradient were the fastest. A stronger reference would shrink the measured economy of the prior; on the cases where the comparison exists, it would not remove it.

Three seeds. Every configuration is repeated on three seeds, which is enough to show that a large effect is not noise and not enough to resolve a small one. On four cases the spread of the reference loop between seeds is itself wider than the two per cent threshold, Section 3.4, and there the number of seeds that reach the target says more than the median. Case 4 is the noisiest: its reference loop ends between seventy and seventy seven across seeds, and one seed never reaches the target of the other two.

Two dimensions, one operating point, one turbulence model. The study is two dimensional, both objectives are single point, and every RANS evaluation uses the Spalart Allmaras model with no transition modelling. That last point bites hardest on case 5, where a fully turbulent computation at a Reynolds number of three hundred thousand is physically questionable, and where the probe carries the low Reynolds regime only at a single, lower Reynolds number, so neither the reference nor the data that judges the members is on firm ground there.

The latent space is a restriction, not a reparametrisation. Eight modes capture ninety nine per cent of the variance of the coefficients but leave about a fifth of the surface spread of the airfoils, and rather more on the designs the free search delivers. The latent does not reproduce the optimum of

the unconstrained search; it finds a different one, and the empirical finding of Chapter 5 is that the different one is better. That is a measured result and not a property of the decomposition.

6.4 Future work

Closer to design practice. Industrial design optimises several operating points at once, in three dimensions, with transition where it matters. The method needs nothing specific to a single point: the anchors of a multi point objective can be ranked as those of a single one. Whether the members still order designs correctly when the objective is a weighted sum of conditions is the first question to answer.

The upper bound of a gate is not measured. Chapter 4 refutes the gate we built, one that reads the uncertainty of a zoo member. An *oracle* gate, trusting on the true error with the hidden evaluation not charged to the budget, would bound from above what any decision rule could earn. Our two runs of it are not conclusive, both stopped at the iteration cap of the harness, and until it is rerun the honest statement is that a perfect gate may buy some anytime advantage while a calibrated one does not.

A selector that commits later, or earlier. On cases 7 and 9 the selector drops a useful prior because six anchors are too few to rank on, and holding it off until more anchors are in deletes the window where the prior helps. A larger initial design, or a prior carried from a similar case already run, are the two ways out, and both cost evaluations that would have to be earned back.

The probe set is a design choice, and everything depends on it. The probe of Section 2.2 was assembled from what was freely available, and every statement this thesis makes about the reliability of a member, about its bias, and about the calibration of its uncertainty is relative to that choice. The next step is to design the probe against the reference actually used, stratified in geometry and in flow condition over the envelope the optimizer visits.

A probe set that the campaigns feed, if it converges. Each optimization run already produces what such a probe needs, designs evaluated with the reference, and the loop already records the residual of every member at every verification. Accumulated over many cases this yields two things at once, a competence map of each member against the reference actually used and a residual model per member calibrated on it, and it costs nothing because the evaluations are paid for anyway. The condition is not a detail. The scheme is a closed loop, and the designs it accumulates are chosen by the optimizer, so they are the population Chapter 4 showed to be selected for the optimism of the surrogate. Enriching on them corrects the model where the optimizer went, which is what one wants, and teaches it the distribution of its own past mistakes, which may not be. Our own attempt is not encouraging: ten successive rounds of between run enrichment on one case moved the delivered design by less than the spread between rounds. The cheapest test of convergence is the one we did not run, hold out a condition entirely, enrich on all the others, and measure whether the correction on the held out condition improves or drifts.

Appendix A

Additional figures

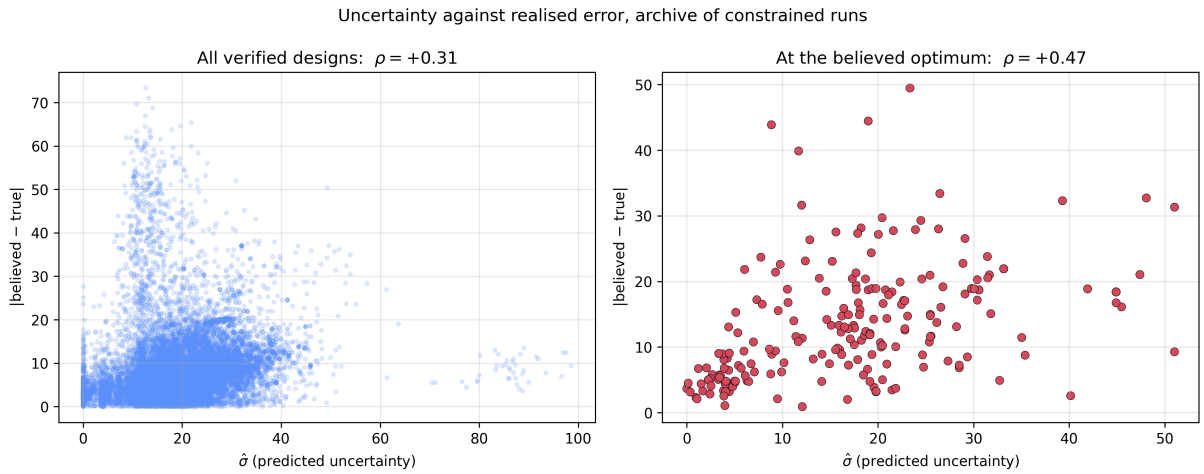


Figure A.1: Predicted uncertainty against realised error on the archive of earlier constrained runs, over all verified designs on the left and at the believed optimum of each run on the right, the rank correlation in each panel. Same behaviour as the controlled experiment of Figure 4.4.

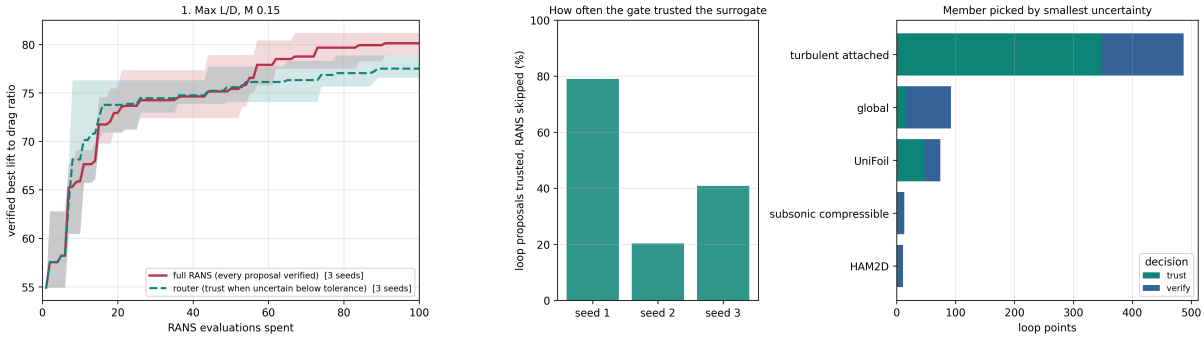


Figure A.2: Router against full RANS on case 1, with the share of proposals trusted per seed and the zoo member picked by the selector at each loop point.

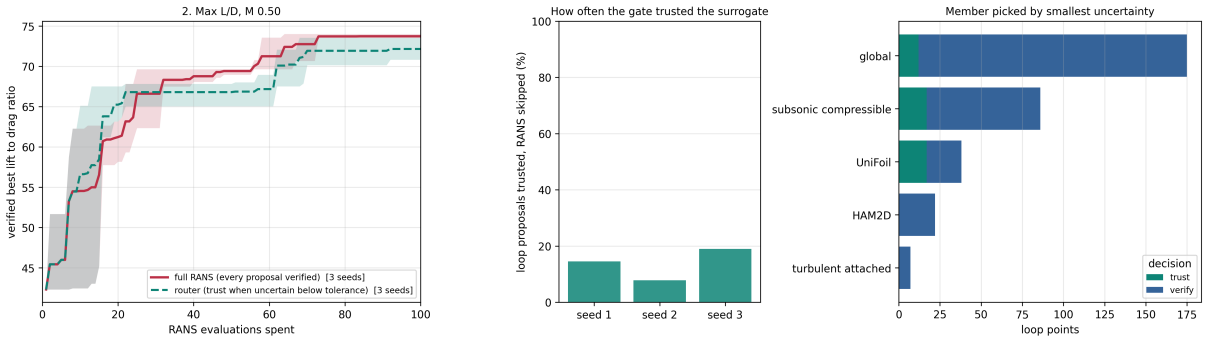


Figure A.3: Same as Figure A.2 for case 2.

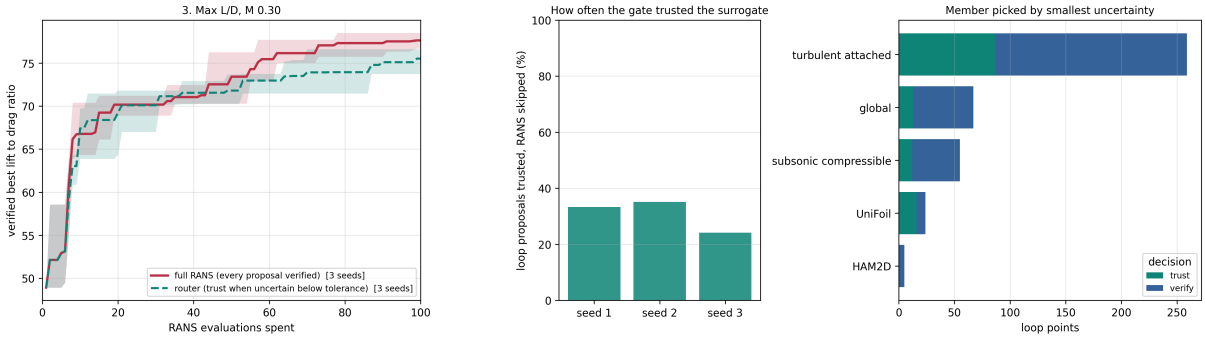


Figure A.4: Same as Figure A.2 for case 3.

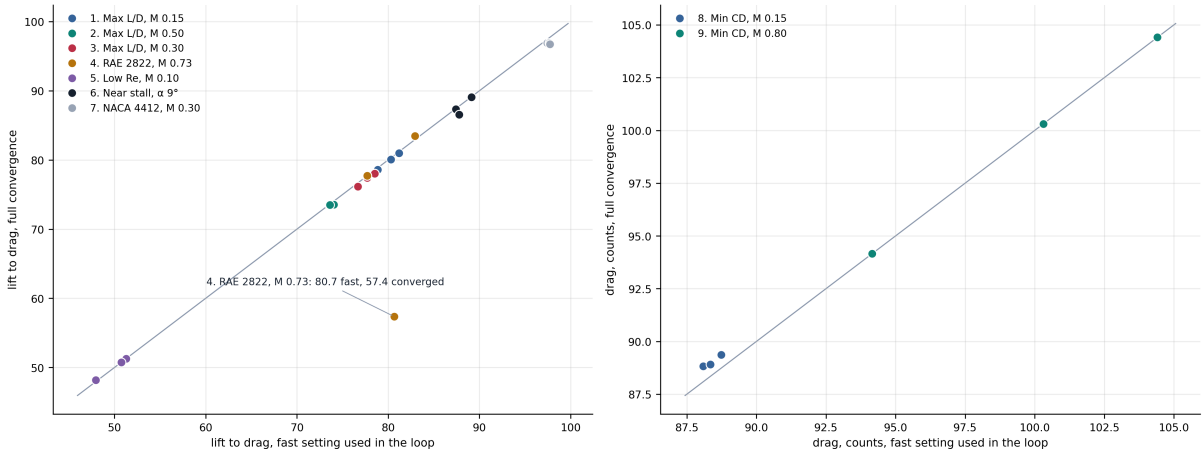


Figure A.5: Fast against fully converged RANS on the delivered designs of the reference loop, nine cases and three seeds, referenced from Section 3.4. Left the lift to drag cases, right the drag cases. The points sit on the diagonal except one transonic run, where the fast setting announces a design that full convergence does not confirm.

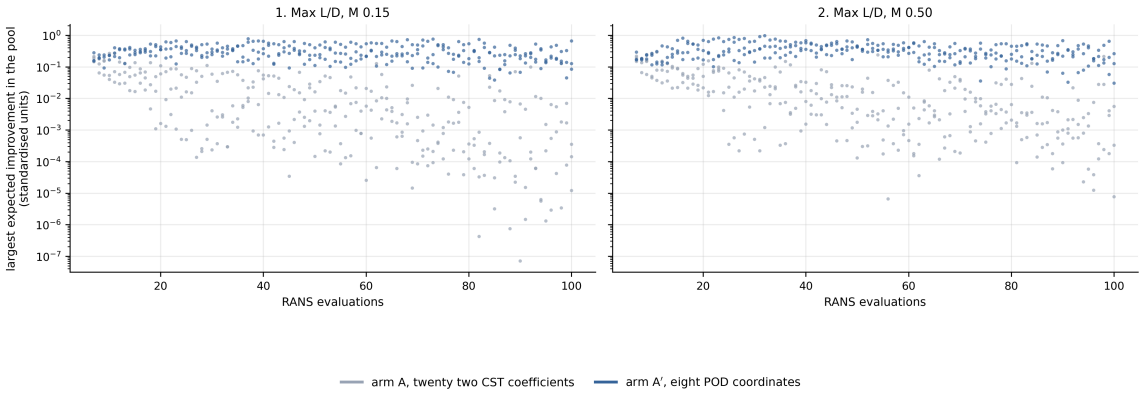


Figure A.6: Largest expected improvement found in the candidate pool at every iteration, arm A in the twenty two CST coefficients against arm A′ in the eight POD coordinates, two cases, three seeds, standardised units of the process on a logarithmic scale. Referenced from Section 5.3.

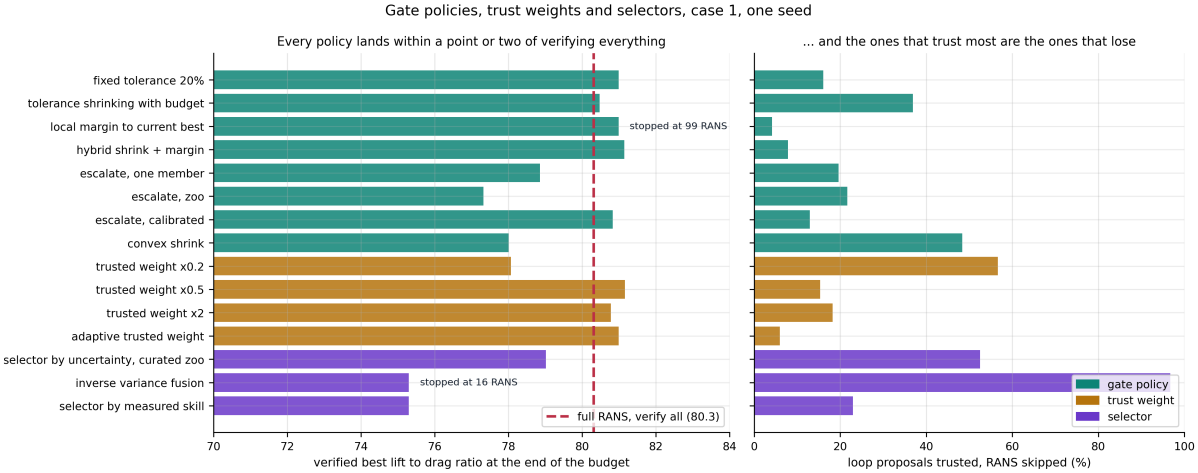


Figure A.7: Every gate policy, trust weight and selector tried on case 1, one seed of an earlier campaign, referenced from Section 4.2. Left, the verified best design at the end of the budget against the full RANS arm in red. Right, the share of loop proposals trusted. The policies that trust little land within a point or two of the full arm, the policies that trust a lot end below it, and none rises above it by more than the spread between seeds.

Appendix B

Admission of the zoo members

The networks of the zoo are admitted as prior candidates only if their predictions are physically possible on the design box of the case. The check costs no RANS evaluation, it reads each member's raw output on a spread population of designs, and its thresholds were fixed before its outcome was looked at. A member is admitted on a case when

- its drag coefficient is strictly positive on at least ninety nine per cent of the designs,
- its lift coefficient stays below three in magnitude on every design,
- on the lift to drag cases, its lift to drag ratio stays below three hundred in magnitude,
- and its prediction is not degenerate: at least ninety per cent of the values are distinct and their spread exceeds one per cent of their median.

The population is the first twenty five verified designs of every run that used no prior, about two hundred and fifty designs per case, which covers the box without having been steered by any member.

Eleven networks were screened, the two external members and nine networks we trained. The seven members of the zoo of Chapter 2 pass on all nine cases. Four regime experts fail, each on physical grounds and by a wide margin, Table B.1. The transonic expert is the one borderline: in its own regime, on the two transonic cases, its drag is positive and it passes, and everywhere else its drag is negative on every design. It was kept out of the zoo on every case so that the zoo is the same throughout the thesis.

Table B.1: The four networks the admission check rejects, and what they return on the design box. The seven members of the zoo pass on every case.

network	fails on	what it returns on the box
transonic expert	7 of 9 cases	drag negative on every design outside the two transonic cases
near stall expert	9 of 9 cases	drag negative on every design, $\lvert C_L \rvert$ up to 4×10^4
post stall expert	9 of 9 cases	drag negative on every design, $\lvert C_L \rvert$ up to 3×10^5, nearly constant
low Reynolds expert	9 of 9 cases	$\lvert C_L \rvert$ up to 4×10^5, 7 to 25 % distinct values

The check matters beyond hygiene. The online selector of Section 5.6 scores a candidate by the number of anchor pairs it orders against the truth, and a pair on which a candidate predicts two equal values is not counted against it. A near constant field is therefore scored as a perfect ranker. On the incompressible minimum drag case the near stall expert returns two distinct values over forty eight anchors, ninety six per cent of its pairs are ties, and a selector given the eleven networks puts its whole weight on it. The admission check removes such a candidate before it can be ranked.

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